

Sine, cosine, tangent and cotangent of an acute angle

Four ratios of sides that depend only on the angle — the whole of trigonometry starts here.

LESSON 19

1 × 40 MIN

GEOMETRY 8, TEMA 15

STAGE 9 · BEYOND

LESSON CLOCK

5'

12'

8'

13'

2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Name the hypotenuse and the sides opposite and adjacent to a given acute angle.
- Define *sin*, *cos*, *tan*, *cot* as ratios of sides.
- Explain why the ratios depend on the angle and not on the size of the triangle.
- Calculate the four ratios from given side lengths.

TEXTBOOK

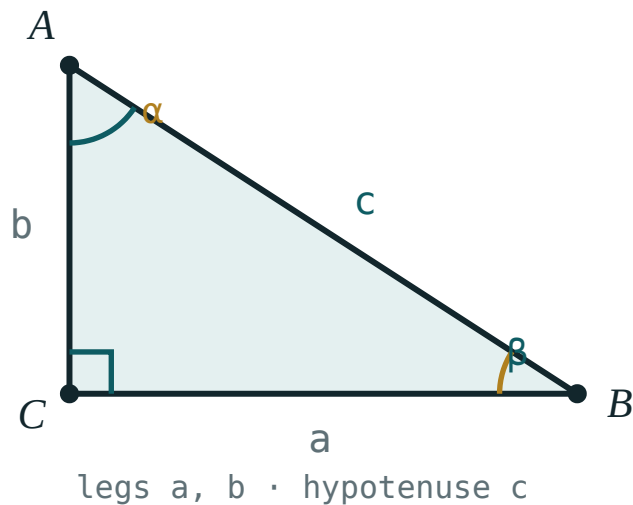
National	Tema 15, pp. 35–37
Cambridge	Extension beyond Stage 9
Workbook	—

01

Explanation

Naming the sides

In a right-angled triangle the side opposite the right angle is the **hypotenuse** — always the longest. The other two are the **legs**.



Right angle at C . The legs are a and b ; the hypotenuse is c . Which leg is “opposite” depends on which acute angle you are looking at.

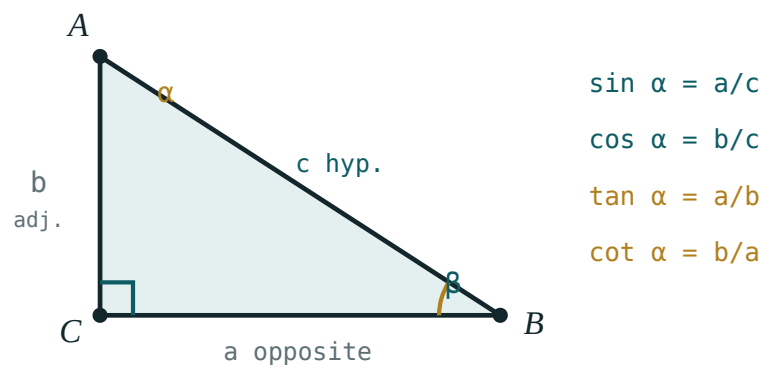
Fix your attention on the acute angle α at A . Then:

- $a = BC$ is the side **opposite** α — it does not touch A ;
- $b = AC$ is the side **adjacent** to α — it does touch A , and it is not the hypotenuse;
- $c = AB$ is the **hypotenuse**.

“OPPOSITE” MOVES

Look at the other acute angle β and the roles swap: b becomes opposite and a becomes adjacent. The hypotenuse never changes.

The four definitions



The four ratios, all measured for the angle α at A .

$$\sin \alpha = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{a}{c} \quad \cos \alpha = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{b}{c}$$

$$\tan \alpha = \frac{\text{opposite}}{\text{adjacent}} = \frac{a}{b} \quad \cot \alpha = \frac{\text{adjacent}}{\text{opposite}} = \frac{b}{a}$$

Two immediate consequences, both worth saying aloud:

- $\tan \alpha \cdot \cot \alpha = 1$ — the two are reciprocals.
- $0 < \sin \alpha < 1$ and $0 < \cos \alpha < 1$, because a leg is always shorter than the hypotenuse. A sine of 1.2 is always an arithmetic error.

Why the ratios depend only on the angle

Draw two right-angled triangles with the same acute angle α . Their third angles are equal too (both $90^\circ - \alpha$), so the triangles are **similar** — one is an enlargement of the other. Enlargement multiplies every side by the same scale factor, and in a ratio of two sides that factor cancels.

$$\frac{a}{c} = \frac{k \cdot a}{k \cdot c}$$

So $\sin \alpha$ is a property of the **angle**, not of the triangle. That is exactly why a single table of values can serve every right-angled triangle in the world.

02 Worked examples

EXAMPLE 1 In a right triangle $a = 3$, $b = 4$, $c = 5$. Find $\sin \alpha$, $\cos \alpha$, $\tan \alpha$ and $\cot \alpha$.

$$\textcircled{1} \quad \sin \alpha = \frac{a}{c} = \frac{3}{5} = 0.6$$

Opposite over hypotenuse.

$$\textcircled{2} \quad \cos \alpha = \frac{b}{c} = \frac{4}{5} = 0.8$$

Adjacent over hypotenuse.

$$\textcircled{3} \quad \tan \alpha = \frac{a}{b} = \frac{3}{4} = 0.75$$

Opposite over adjacent.

$$\textcircled{4} \quad \cot \alpha = \frac{b}{a} = \frac{4}{3}$$

The reciprocal of the tangent.

ANSWER

$0.6, 0.8, 0.75, \frac{4}{3}$

EXAMPLE 2 *In a right triangle $\sin \alpha = \frac{5}{13}$. Find $\cos \alpha$ and $\tan \alpha$.*

① *opposite = 5, hypotenuse = 13*
Read the ratio as two side lengths.

② $b = \sqrt{13^2 - 5^2} = \sqrt{144} = 12$
Pythagoras gives the third side.

③ $\cos \alpha = \frac{12}{13}$

④ $\tan \alpha = \frac{5}{12}$

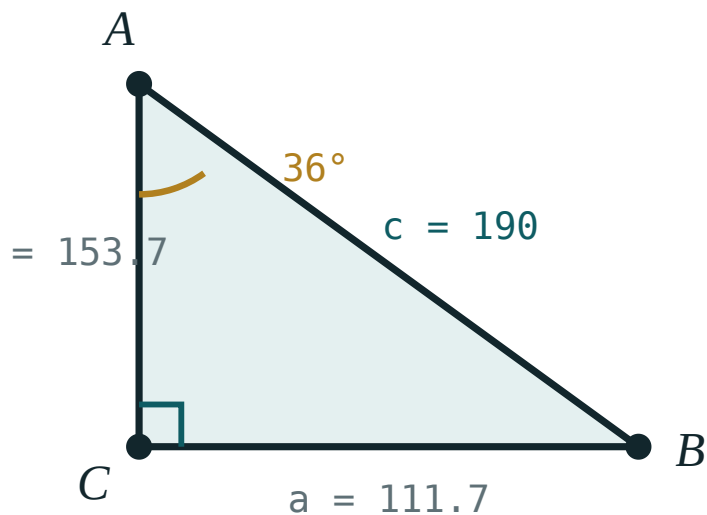
ANSWER

$$\cos \alpha = \frac{12}{13}, \tan \alpha = \frac{5}{12}$$

03 Interactive model

Move the angle slider and watch the ratios change while the shape stays a right triangle.

Solve the right triangle

 α 36° β 54° a (opposite) **111.7**b (adjacent) **153.7**c (hyp) **190** $\sin \alpha$ **0.59** $\cos \alpha$ **0.81** $\tan \alpha$ **0.73** $a^2 + b^2$ **36100** c^2 **36100****04 Terminology**

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Right-angled triangle	To'g'ri burchakli uchburchak	Прямоугольный треугольник
Hypotenuse	Gipotenuza	Гипотенуза
Leg (cathetus)	Katet	Катет
Opposite side	Qarshi yotgan katet	Противолежащий катет
Adjacent side	Yondosh katet	Прилежащий катет
Sine	Sinus	Синус
Cosine	Kosinus	Косинус
Tangent	Tangens	Тангенс
Cotangent	Kotangens	Котангенс
Ratio	Nisbat	Отношение
Acute angle	O'tkir burchak	Острый угол

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$\sin \alpha$ is:

$\frac{\text{adjacent}}{\text{hypotenuse}}$

$\frac{\text{opposite}}{\text{hypotenuse}}$

$\frac{\text{opposite}}{\text{adjacent}}$

$\frac{\text{hypotenuse}}{\text{opposite}}$

For an acute angle, $\sin \alpha$ is always:

greater than 1

between 0 and 1

equal to 1

negative

$\tan \alpha \cdot \cot \alpha$ equals:

0

1

2

$\sin \alpha$

Two right triangles have the same acute angle. Their sines are:

different

equal

reciprocal

not comparable

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $a = 3, c = 5$. Find $\sin \alpha$.

ANSWER 0.6

2. $b = 4, c = 5$. Find $\cos \alpha$.

ANSWER 0.8

3. $a = 3, b = 4$. Find $\tan \alpha$.

ANSWER 0.75

4. $a = 3, b = 4$. Find $\cot \alpha$.

ANSWER $\frac{4}{3}$

5. Which side is opposite the right angle?

ANSWER The hypotenuse.

6. Can $\sin \alpha = 1.2$ for an acute angle?

ANSWER No — the sine of an acute angle is less than 1.

7. $a = 6, c = 10$. Find $\sin \alpha$.

ANSWER 0.6

Medium · the standard question

7 PROBLEMS

1. $a = 5, b = 12, c = 13$. Find all four ratios for α .

ANSWER $\sin = \frac{5}{13}, \cos = \frac{12}{13}, \tan = \frac{5}{12}, \cot = \frac{12}{5}$

2. $\sin \alpha = \frac{5}{13}$. Find $\cos \alpha$.

ANSWER $\frac{12}{13}$

3. $\cos \alpha = \frac{8}{17}$. Find $\sin \alpha$.

ANSWER $\frac{15}{17}$

4. $\tan \alpha = \frac{3}{4}$. Find $\cot \alpha$.

ANSWER $\frac{4}{3}$

5. $a = 8, c = 17$. Find b and $\cos \alpha$.

ANSWER $b = 15, \cos \alpha = \frac{15}{17}$

6. In the same triangle, write $\sin \beta$ in terms of the sides.

ANSWER $\sin \beta = \frac{b}{c}$

7. $a = 9, b = 12$. Find c and $\sin \alpha$.

ANSWER $c = 15, \sin \alpha = 0.6$

Hard · stretch and reason

7 PROBLEMS

1. $\sin \alpha = \frac{7}{25}$. Find $\cos \alpha$, $\tan \alpha$ and $\cot \alpha$.

ANSWER $\cos = \frac{24}{25}$, $\tan = \frac{7}{24}$, $\cot = \frac{24}{7}$

2. $\tan \alpha = 1$. What is α , and what are $\sin \alpha$ and $\cos \alpha$?

ANSWER $\alpha = 45^\circ$, both ratios $\frac{\sqrt{2}}{2}$

3. Show that $\sin \alpha = \cos \beta$ in a right triangle with acute angles α and β .

ANSWER Both equal $\frac{a}{c}$ for the same leg, since the two angles are complementary.

4. A right triangle has legs in the ratio 1 : 2. Find $\tan \alpha$ for the smaller angle.

ANSWER $\tan \alpha = \frac{1}{2} = 0.5$

5. $c = 20$ and $\sin \alpha = 0.6$. Find both legs.

ANSWER $a = 12$, $b = 16$

6. Prove that $\tan \alpha = \frac{\sin \alpha}{\cos \alpha}$

ANSWER $\frac{a/c}{b/c} = \frac{a}{b} = \tan \alpha$

7. A right triangle has $\cot \alpha = 2.4$. Find $\tan \alpha$ and the ratio of the legs.

ANSWER $\tan \alpha = \frac{5}{12}$; legs in the ratio 5 : 12

07 Homework

Homework — 5 problems

Geometry 8, Tema 15, pp. 35–37. Draw and label a triangle for every question.

1. $a = 7, b = 24, c = 25$. Find the four ratios for α .
2. $\sin \alpha = 0.8$. Find $\cos \alpha$ and $\tan \alpha$.
3. $\cos \alpha = \frac{20}{29}$. Find $\sin \alpha$.
4. $a = 15, c = 17$. Find b , then $\tan \alpha$.
5. Explain in two sentences why $\sin \alpha$ does not depend on the size of the triangle.

Pythagoras’ theorem, and different proofs of it

The most famous relation in mathematics, and three ways of seeing why it must be true.

LESSON 20–21

2 × 40 MIN

GEOMETRY 8, TEMA 17

STAGE 9 · 5.5

LESSON CLOCK

5'

12'

8'

13'

2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- State Pythagoras’ theorem precisely.
- Prove it by the square-rearrangement method.
- Find the hypotenuse or a leg from the other two sides.
- Recognise the common Pythagorean triples.

TEXTBOOK

National	Tema 17, pp. 41–43
Cambridge	Learner’s Book pp. 124–126
Workbook	Workbook 5.5

01

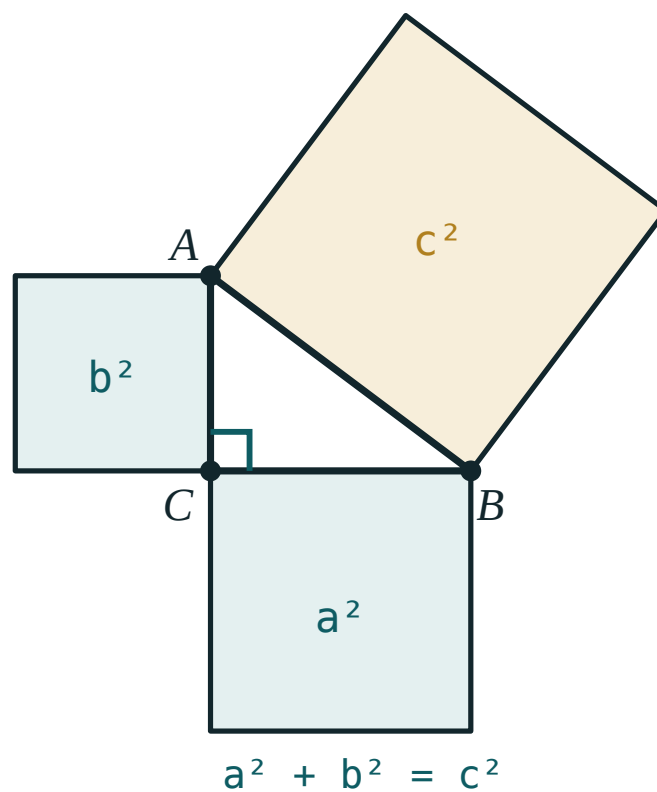
Explanation

The theorem

THEOREM

In a right-angled triangle the square of the hypotenuse equals the sum of the squares of the legs:

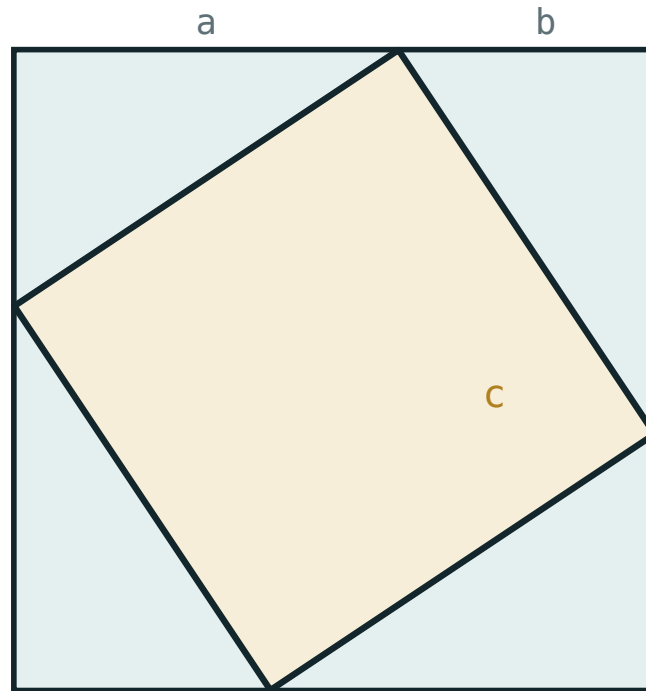
$$a^2 + b^2 = c^2$$



The two smaller squares together have exactly the area of the large one.

The picture is the statement: build a square on each side, and the two small ones together fill the big one exactly.

Proof 1 — the rearrangement



$$(a + b)^2 = 4 \cdot \frac{1}{2}ab + c^2$$

Four copies of the triangle inside a square of side $a + b$. What is left in the middle is a square of side c .

Take a square of side $a + b$ and place four copies of the right triangle inside it, one in each corner. The area can be counted two ways:

$$(a + b)^2 = 4 \cdot \frac{1}{2}ab + c^2$$

$$a^2 + 2ab + b^2 = 2ab + c^2$$

$$a^2 + b^2 = c^2 \blacksquare$$

The middle really is a square: each of its angles is $180^\circ - \alpha - \beta = 180^\circ - 90^\circ = 90^\circ$, and all four sides are hypotenuses of congruent triangles.

Proof 2 — by similar triangles

Drop the height CH from the right angle onto the hypotenuse. It cuts the triangle into two smaller triangles, each similar to the original (they share an acute angle and have a right angle). Comparing sides gives $a^2 = c \cdot BH$ and $b^2 = c \cdot AH$. Adding:

$$a^2 + b^2 = c(BH + AH) = c \cdot c = c^2 \blacksquare$$

This proof is shorter, and it produces two useful formulas on the way.

Using the theorem

To find the **hypotenuse**, add and take the root. To find a **leg**, subtract first:

$$c = \sqrt{a^2 + b^2} \quad a = \sqrt{c^2 - b^2}$$

TRIPLES WORTH RECOGNISING ON SIGHT

3, 4, 5 · 5, 12, 13 · 8, 15, 17 · 7, 24, 25 · 20, 21, 29 — and every multiple of them, such as 6, 8, 10 and 9, 12, 15.

THE HYPOTENUSE IS THE LONGEST SIDE

If your answer for c comes out smaller than a leg, you have added where you should have subtracted.

02 Worked examples

EXAMPLE 1 The legs of a right triangle are 9 cm and 12 cm. Find the hypotenuse.

① $c^2 = 9^2 + 12^2$
Pythagoras.

② $= 81 + 144 = 225$

③ $c = \sqrt{225} = 15 \text{ cm}$

Recognise 9, 12, 15 — three times 3, 4, 5.

ANSWER

15 cm

EXAMPLE 2 The hypotenuse is 26 cm and one leg is 10 cm. Find the other leg.

① $b^2 = 26^2 - 10^2$

Looking for a leg, so subtract.

② $= 676 - 100 = 576$

③ $b = 24 \text{ cm}$

The triple 10, 24, 26 is twice 5, 12, 13.

ANSWER

24 cm

EXAMPLE 3 A ladder 5 m long leans against a wall with its foot 3 m from the wall. How high does it reach?

① The wall, the ground and the ladder form a right triangle.

The ladder is the hypotenuse.

② $h^2 = 5^2 - 3^2 = 25 - 9 = 16$

③ $h = 4 \text{ m}$

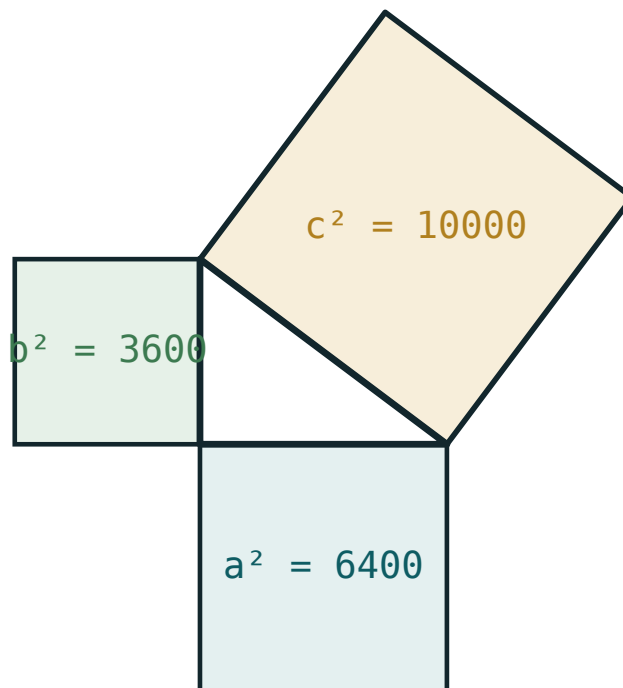
ANSWER

4 m

03 Interactive model

Change both legs and watch $a^2 + b^2$ and c^2 stay equal to the last digit.

$$a^2 + b^2 = c^2$$



a 80

b 60

c 100

a^2 6400

b^2 3600

$a^2 + b^2$ 10000

c^2 10000

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Pythagoras' theorem	Pifagor teoremasi	Теорема Пифагора
Hypotenuse	Gipotenuza	Гипотенуза
Leg (cathetus)	Katet	Катет
Square of a number	Sonning kvadrati	Квадрат числа
Area	Yuza	Площадь
Proof	Isbot	Доказательство
Rearrangement	Qayta joylashtirish	Перекладывание
Pythagorean triple	Pifagor uchligi	Пифагорова тройка
Right angle	To'g'ri burchak	Прямой угол

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Pythagoras' theorem says:

$$a + b = c$$

$$a^2 + b^2 = c^2$$

$$a^2 - b^2 = c^2$$

$$ab = c^2$$

Legs 6 and 8. The hypotenuse is:

$$10$$

$$14$$

$$48$$

$$7$$

Hypotenuse 13, one leg 5. The other leg is:

$$8$$

$$12$$

$$18$$

$$14$$

Which is a Pythagorean triple?

2, 3, 4

5, 12, 13

4, 5, 6

6, 7, 8

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. *Legs 3 and 4. Find the hypotenuse.*

ANSWER 5

2. *Legs 6 and 8. Find the hypotenuse.*

ANSWER 10

3. *Legs 5 and 12. Find the hypotenuse.*

ANSWER 13

4. *Hypotenuse 5, one leg 3. Find the other leg.*

ANSWER 4

5. *Hypotenuse 13, one leg 12. Find the other leg.*

ANSWER 5

6. *Legs 9 and 12. Find the hypotenuse.*

ANSWER 15

7. *Is 3, 4, 5 a Pythagorean triple?*

ANSWER Yes — $9 + 16 = 25$.

Medium · the standard question

7 PROBLEMS

1. *Legs 8 and 15. Find the hypotenuse.*

ANSWER 17

2. *Legs 7 and 24. Find the hypotenuse.*

ANSWER 25

3. *Hypotenuse 26, one leg 10. Find the other leg.*

ANSWER 24

4. *Hypotenuse 17, one leg 8. Find the other leg.*

ANSWER 15

5. *A ladder 5 m long has its foot 3 m from a wall. How high does it reach?*

ANSWER 4 m

6. *A rectangle is 9 cm by 12 cm. Find its diagonal.*

ANSWER 15 cm

7. *Legs 1 and 1. Find the hypotenuse.*

ANSWER $\sqrt{2}$

Hard · stretch and reason

7 PROBLEMS

1. An isosceles triangle has equal sides 13 and base 10. Find its height.

ANSWER $h = \sqrt{169 - 25} = 12$

2. An equilateral triangle has side 6. Find its height.

ANSWER $h = \sqrt{36 - 9} = 3\sqrt{3}$

3. A rhombus has diagonals 16 and 30. Find its side.

ANSWER $\sqrt{64 + 225} = 17$

4. A right triangle has hypotenuse 20 and one leg 12. Find its area.

ANSWER other leg 16, area 96

5. A ladder 13 m long reaches 12 m up a wall. How far is its foot from the wall?

ANSWER 5 m

6. A trapezium has bases 6 and 16 and equal legs 13. Find its height.

ANSWER $\frac{16 - 6}{2} = 5$, so $h = 12$

7. Complete the proof: the middle shape in the rearrangement is a square. Why are its angles 90° ?

ANSWER Each is $180^\circ - \alpha - \beta$, and $\alpha + \beta = 90^\circ$ in the right triangle.

07 Homework

Homework — 6 problems

Geometry 8, Tema 17, pp. 41–43. Draw a labelled figure for questions 4–6.

- Legs 20 and 21. Find the hypotenuse.
- Hypotenuse 25, one leg 7. Find the other leg.
- A rectangle is 8 cm by 15 cm. Find its diagonal.

4. *An isosceles triangle has equal sides 10 and base 12. Find its height.*
5. *A ladder 10 m long has its foot 6 m from a wall. How high does it reach?*
6. *Write out the rearrangement proof of Pythagoras' theorem with a labelled figure.*

The converse of Pythagoras’ theorem, and applications

Running the theorem backwards to test for a right angle — and using it on real shapes.

LESSON 22–23

2 × 40 MIN

GEOMETRY 8, ТЕМЫ 18–19

STAGE 9 · 5.5

LESSON CLOCK

5'

12'

8'

13'

2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- State the converse of Pythagoras’ theorem and use it to test a triangle.
- Classify a triangle as acute, right or obtuse from its three sides.
- Apply Pythagoras to rectangles, rhombuses, trapeziums and isosceles triangles.
- Solve practical distance problems.

TEXTBOOK

National	Темы 18–19, pp. 44–48
Cambridge	Learner’s Book pp. 124–126
Workbook	Workbook 5.5

01

Explanation

The converse

CONVERSE THEOREM

If the sides of a triangle satisfy $a^2 + b^2 = c^2$, where c is the longest side, then the triangle is **right-angled**, with the right angle opposite c .

The original theorem says “right angle $\implies$ the relation”. The converse says “the relation $\implies$ a right angle”. Both are true here, but that is a piece of luck, not a rule — most theorems have a false converse, which is exactly why the two directions get separate names.

How to use it. Square all three sides. If the two smaller squares add to the largest, the triangle is right-angled.

Classifying any triangle

Comparing $a^2 + b^2$ with c^2 tells you more than just “right or not”:

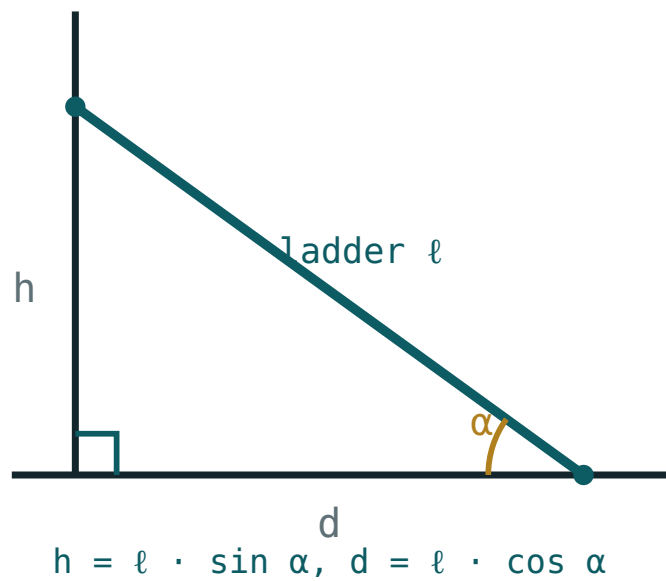
COMPARISON	THE TRIANGLE IS	EXAMPLE
$a^2 + b^2 = c^2$	right-angled	3, 4, 5
$a^2 + b^2 > c^2$	acute-angled	4, 5, 6
$a^2 + b^2 < c^2$	obtuse-angled	2, 3, 4

ALWAYS TEST WITH THE LONGEST SIDE AS C

For 5, 12, 13 the test is $25 + 144 = 169 \checkmark$. Testing $25 + 169$ against 144 proves nothing.

The four standard applications

- **Diagonal of a rectangle.** $d = \sqrt{a^2 + b^2}$.
- **Height of an isosceles triangle.** Drop the height to the base; it bisects the base, so $h = \sqrt{\text{side}^2 - (\text{base}/2)^2}$.
- **Side of a rhombus.** The half-diagonals are perpendicular legs: $a = \sqrt{(d_1/2)^2 + (d_2/2)^2}$.
- **Height of an isosceles trapezium.** Each end triangle has horizontal leg $\frac{a - b}{2}$.



The ladder problem: the wall, the ground and the ladder always form a right triangle.

The practical version is always the same picture — a vertical, a horizontal and a slant.

02 Worked examples

EXAMPLE 1 Is the triangle with sides 8, 15, 17 right-angled?

- 1 *longest side $c = 17$*
Identify it first.

- 2 $8^2 + 15^2 = 64 + 225 = 289$

- 3 $17^2 = 289$

- 4 They are equal, so by the converse the triangle is right-angled.
The right angle is opposite the side 17.

ANSWER

Yes — right-angled.

EXAMPLE 2 Classify the triangle with sides 4, 5, 6.

- 1 $c = 6$
The longest side.

- 2 $4^2 + 5^2 = 16 + 25 = 41$

- 3 $6^2 = 36$

- 4 $41 > 36$
The sum of the smaller squares is larger.

ANSWER

Acute-angled.

EXAMPLE 3 A rhombus has diagonals 10 cm and 24 cm. Find its side and perimeter.

① *half-diagonals 5 and 12*

The diagonals bisect each other.

② They are perpendicular, so they are the legs of a right triangle.

The side of the rhombus is the hypotenuse.

③ $a = \sqrt{25 + 144} = 13 \text{ cm}$

④ $P = 4 \cdot 13 = 52 \text{ cm}$

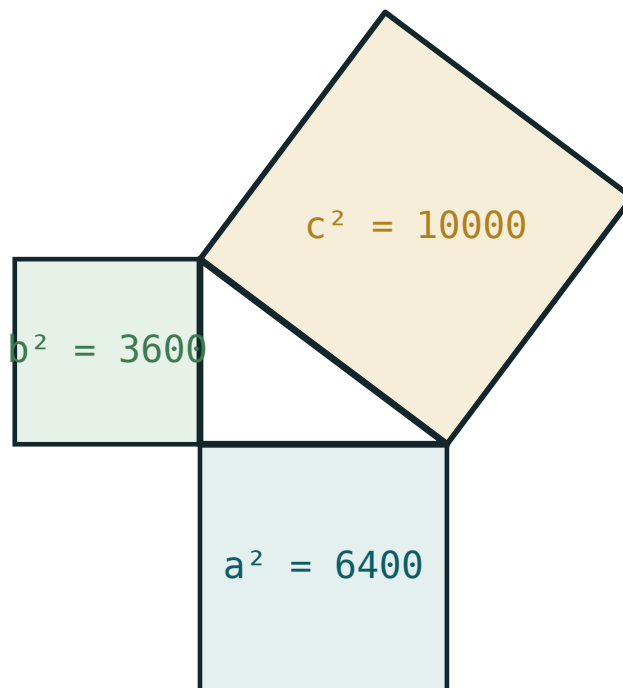
ANSWER

$a = 13 \text{ cm}, P = 52 \text{ cm}$

03 Interactive model

Set the legs to 8 and 15 and read c — then ask the class to name the triple.

Test a triangle for a right angle



a 80

b 60

c 100

a^2 6400

b^2 3600

$a^2 + b^2$ 10000

c^2 10000

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Converse theorem	Teskari teorema	Обратная теорема
Test for a right angle	To'g'ri burchak alomati	Признак прямого угла
Acute triangle	O'tkir burchakli uchburchak	Остроугольный треугольник
Obtuse triangle	O'tmas burchakli uchburchak	Тупоугольный треугольник
Longest side	Eng uzun tomon	Наибольшая сторона
Diagonal	Diagonal	Диагональ
Height	Balandlik	Высота
Distance	Masofa	Расстояние
Apply	Qo'llash	Применить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Sides 9, 40, 41. The triangle is:

right-angled

acute

obtuse

impossible

Sides 2, 3, 4. The triangle is:

right-angled

acute

obtuse

impossible

To test sides 7, 24, 25 you compare:

$7^2 + 25^2$ with 24^2

$7^2 + 24^2$ with 25^2

$24^2 + 25^2$ with 7^2

any of these

A rhombus with diagonals 6 and 8 has side:

5

7

10

14

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. *Is 3, 4, 5 right-angled?*

ANSWER Yes.

2. *Is 5, 12, 13 right-angled?*

ANSWER Yes.

3. *Is 4, 5, 6 right-angled?*

ANSWER No — it is acute.

4. *Is 2, 3, 4 right-angled?*

ANSWER No — it is obtuse.

5. *A rectangle is 6 by 8. Find its diagonal.*

ANSWER 10

6. *A rhombus has diagonals 6 and 8. Find its side.*

ANSWER 5

7. *Which side must be squared alone in the test?*

ANSWER The longest one.

Medium · the standard question

7 PROBLEMS

1. *Is 8, 15, 17 right-angled?*

ANSWER Yes.

2. *Classify the triangle with sides 4, 5, 6.*

ANSWER Acute.

3. *Classify the triangle with sides 5, 6, 10.*

ANSWER $25 + 36 = 61 < 100$ — obtuse.

4. *A rhombus has diagonals 10 and 24. Find its side and perimeter.*

ANSWER 13, 52

5. *An isosceles triangle has equal sides 10 and base 16. Find its height.*

ANSWER 6

6. *A rectangle has diagonal 25 and one side 7. Find the other side.*

ANSWER 24

7. *A ladder 17 m long has its foot 8 m from a wall. How high does it reach?*

ANSWER 15 m

Hard · stretch and reason

7 PROBLEMS

1. An isosceles trapezium has bases 8 and 20 and legs 10. Find its height and area.

ANSWER $\frac{20-8}{2} = 6, h = 8, S = 112$

2. An equilateral triangle has side 10. Find its height and area.

ANSWER $h = 5\sqrt{3}, S = 25\sqrt{3}$

3. A right triangle has perimeter 30 and hypotenuse 13. Find its legs.

ANSWER $a + b = 17$ and $a^2 + b^2 = 169$ give $ab = 60$: legs 5 and 12.

4. Two points are 3 km east and 4 km north of a village. How far apart are they?

ANSWER 5 km

5. Show that 20, 21, 29 is a Pythagorean triple.

ANSWER $400 + 441 = 841 = 29^2$

6. A rhombus has side 25 and one diagonal 14. Find the other diagonal.

ANSWER half-diagonal $\sqrt{625 - 49} = 24$, so the diagonal is 48.

7. For which x is the triangle with sides 6, 8, x right-angled?

ANSWER $x = 10$ (hypotenuse) or $x = 2\sqrt{7}$ (leg).

07 Homework

Homework — 6 problems

Geometry 8, Темы 18–19, pp. 44–48.

1. Is 9, 12, 15 right-angled? Show the test.
2. Classify the triangle with sides 7, 8, 12.
3. A rhombus has diagonals 16 and 30. Find its side and perimeter.
4. An isosceles triangle has equal sides 17 and base 16. Find its height.

5. *A rectangle has diagonal 41 and one side 9. Find the other side and the area.*
6. *A ladder 25 m long reaches 24 m up a wall. How far is its foot from the wall?*

The basic trigonometric identity

Pythagoras’ theorem, divided through by c^2 — and suddenly it is a statement about angles.

LESSON 24

1 × 40 MIN

GEOMETRY 8, ТЕМЫ 20–21

STAGE 9 · BEYOND

LESSON CLOCK

5'

12'

8'

13'

2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Derive $\sin^2\alpha + \cos^2\alpha = 1$ from Pythagoras' theorem.
- Find one ratio from another using the identity.
- Use the corollaries $\tan \alpha = \sin \alpha / \cos \alpha$ and $\tan \alpha \cdot \cot \alpha = 1$.
- Choose the right sign — for an acute angle every ratio is positive.

TEXTBOOK

National	Темы 20–21, pp. 49–51
Cambridge	Extension beyond Stage 9
Workbook	—

01

Explanation

The derivation

Start from Pythagoras in the triangle of the last lesson and divide every term by c^2 :

$$a^2 + b^2 = c^2$$

$$\frac{a^2}{c^2} + \frac{b^2}{c^2} = 1$$

$$\left(\frac{a}{c}\right)^2 + \left(\frac{b}{c}\right)^2 = 1$$

THE BASIC TRIGONOMETRIC IDENTITY

$$\sin^2\alpha + \cos^2\alpha = 1$$

true for **every** acute angle α .

Notation: $\sin^2 \alpha$ means $(\sin \alpha)^2$, never $\sin(\alpha^2)$.

Three corollaries

$$\sin \alpha = \sqrt{1 - \cos^2 \alpha} \quad \cdot \quad \cos \alpha = \sqrt{1 - \sin^2 \alpha}$$

The positive root is the right one every time, because for an acute angle all four ratios are positive.

$$\tan \alpha = \frac{\sin \alpha}{\cos \alpha} \quad \cdot \quad \cot \alpha = \frac{\cos \alpha}{\sin \alpha} \quad \cdot \quad \tan \alpha \cdot \cot \alpha = 1$$

Proof of the first: $\frac{\sin \alpha}{\cos \alpha} = \frac{a/c}{b/c} = \frac{a}{b} = \tan \alpha$. The c cancels.

Using it

Given one ratio, the identity produces all the others without ever drawing a triangle:

$$\sin \alpha = 0.6 \implies \cos^2 \alpha = 1 - 0.36 = 0.64 \implies \cos \alpha = 0.8$$

and then $\tan \alpha = \frac{0.6}{0.8} = 0.75$.

A QUICK SANITY CHECK

Both $\sin \alpha$ and $\cos \alpha$ must lie strictly between 0 and 1, and $\sin^2 \alpha + \cos^2 \alpha$ must come to exactly 1. If it does not, go back.

02 Worked examples

EXAMPLE 1 Given $\sin \alpha = \frac{3}{5}$, find $\cos \alpha$ and $\tan \alpha$.

$$(1) \quad \cos^2 \alpha = 1 - \left(\frac{3}{5}\right)^2 = 1 - \frac{9}{25} = \frac{16}{25}$$

The identity.

$$(2) \quad \cos \alpha = \frac{4}{5}$$

Positive root — the angle is acute.

$$(3) \quad \tan \alpha = \frac{\sin \alpha}{\cos \alpha} = \frac{3/5}{4/5} = \frac{3}{4}$$

ANSWER

$$\cos \alpha = 0.8, \tan \alpha = 0.75$$

EXAMPLE 2 Given $\cos \alpha = \frac{5}{13}$, find $\sin \alpha$ and $\cot \alpha$.

$$\textcircled{1} \quad \sin^2 \alpha = 1 - \frac{25}{169} = \frac{144}{169}$$

$$\textcircled{2} \quad \sin \alpha = \frac{12}{13}$$

$$\textcircled{3} \quad \cot \alpha = \frac{\cos \alpha}{\sin \alpha} = \frac{5}{12}$$

ANSWER

$$\sin \alpha = \frac{12}{13}, \cot \alpha = \frac{5}{12}$$

EXAMPLE 3 Simplify $(1 - \sin \alpha)(1 + \sin \alpha)$

$$\textcircled{1} \quad = 1 - \sin^2 \alpha$$

Difference of two squares.

$$\textcircled{2} \quad \sin^2 \alpha + \cos^2 \alpha = 1$$

so $1 - \sin^2 \alpha = \cos^2 \alpha$

$$\textcircled{3} \quad = \cos^2 \alpha$$

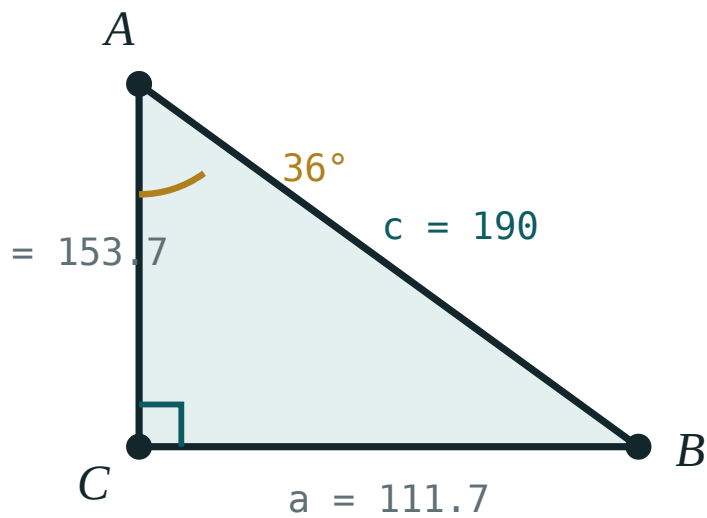
ANSWER

$$\cos^2 \alpha$$

03 Interactive model

Move the angle and watch $\sin^2 + \cos^2$ stay at 1.00 for every position.

sin, cos and the identity

 α 36° β 54° a (opposite) **111.7**b (adjacent) **153.7**c (hyp) **190** $\sin \alpha$ **0.59** $\cos \alpha$ **0.81** $\tan \alpha$ **0.73** $a^2 + b^2$ **36100** c^2 **36100****04 Terminology**

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Identity	Ayniyat	Тождество
Basic trigonometric identity	Asosiy trigonometrik ayniyat	Основное тригонометрическое тождество
Corollary	Natija	Следствие
Square of the sine	Sinusning kvadrati	Квадрат синуса
Derive	Keltirib chiqarish	Вывести
Divide through	Hadma-had bo'lish	Разделить почленно
Positive	Musbat	Положительный
Substitute	O'rniga qo'yish	Подставить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$\sin^2 \alpha + \cos^2 \alpha$ equals:

0

1

2

$\tan \alpha$

If $\sin \alpha = 0.6$ then $\cos \alpha$ is:

0.4

0.8

0.36

1.4

$\tan \alpha$ equals:

$\frac{\cos \alpha}{\sin \alpha}$

$\frac{\sin \alpha}{\cos \alpha}$

$\sin \alpha \cdot \cos \alpha$

$1 - \cos \alpha$

$1 - \sin^2 \alpha$ equals:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\sin \alpha = 0.6$. Find $\cos \alpha$.

ANSWER 0.8

2. $\cos \alpha = 0.8$. Find $\sin \alpha$.

ANSWER 0.6

3. $\sin \alpha = \frac{3}{5}$. Find $\cos \alpha$.

ANSWER $\frac{4}{5}$

4. $\cos \alpha = \frac{12}{13}$. Find $\sin \alpha$.

ANSWER $\frac{5}{13}$

5. Simplify $1 - \cos^2 \alpha$

ANSWER $\sin^2 \alpha$

6. Simplify $1 - \sin^2 \alpha$

ANSWER $\cos^2 \alpha$

7. $\tan \alpha \cdot \cot \alpha = ?$

ANSWER 1

Medium · the standard question

7 PROBLEMS

1. $\sin \alpha = \frac{3}{5}$. Find $\cos \alpha$ and $\tan \alpha$.

ANSWER $\frac{4}{5}$, $\frac{3}{4}$

2. $\cos \alpha = \frac{5}{13}$. Find $\sin \alpha$ and $\cot \alpha$.

ANSWER $\frac{12}{13}$, $\frac{5}{12}$

3. $\sin \alpha = \frac{8}{17}$. Find $\cos \alpha$.

ANSWER $\frac{15}{17}$

4. Simplify $(1 - \sin \alpha)(1 + \sin \alpha)$

ANSWER $\cos^2 \alpha$

5. Simplify $\sin^2 \alpha - 1$

ANSWER $-\cos^2 \alpha$

6. $\tan \alpha = \frac{3}{4}$. Find $\cot \alpha$.

ANSWER $\frac{4}{3}$

7. Simplify $\frac{\sin \alpha}{\cos \alpha} \cdot \cot \alpha$

ANSWER 1

Hard · stretch and reason

7 PROBLEMS

1. $\sin \alpha = \frac{7}{25}$. Find $\cos \alpha$, $\tan \alpha$ and $\cot \alpha$.

ANSWER $\frac{24}{25}$, $\frac{7}{24}$, $\frac{24}{7}$

2. Simplify $\sin^2 \alpha + \cos^2 \alpha + \tan \alpha \cdot \cot \alpha$

ANSWER $1 + 1 = 2$

3. Simplify $(\sin \alpha + \cos \alpha)^2$

ANSWER $1 + 2 \sin \alpha \cos \alpha$

4. Simplify $(\sin \alpha + \cos \alpha)^2 + (\sin \alpha - \cos \alpha)^2$

ANSWER 2

5. Prove that $\frac{1}{\cos^2 \alpha} = 1 + \tan^2 \alpha$

ANSWER Divide the basic identity by $\cos^2 \alpha$.

6. $\tan \alpha = 2$. Find $\sin \alpha$ and $\cos \alpha$.

ANSWER $\cos^2 \alpha = \frac{1}{5}$, so $\cos \alpha = \frac{1}{\sqrt{5}}$ and $\sin \alpha = \frac{2}{\sqrt{5}}$

7. Simplify $\cos^2 \alpha - \sin^2 \alpha + 2 \sin^2 \alpha$

ANSWER 1

07 Homework

Homework — 5 problems

Geometry 8, Темы 20–21, pp. 49–51.

1. $\sin \alpha = \frac{15}{17}$. Find $\cos \alpha$ and $\tan \alpha$.

2. $\cos \alpha = 0.28$. Find $\sin \alpha$.
3. Simplify $(1 - \cos \alpha)(1 + \cos \alpha)$
4. Simplify $\sin^2 \alpha \cdot \cot \alpha \cdot \tan \alpha$
5. Derive $\sin^2 \alpha + \cos^2 \alpha = 1$ from Pythagoras' theorem, with a labelled figure.

Trigonometric functions of complementary angles

The two acute angles of a right triangle add to 90° — and that single fact swaps sine with cosine.

LESSON 25

1 × 40 MIN

GEOMETRY 8, TEMA 22

STAGE 9 · BEYOND

LESSON CLOCK

5'

12'

8'

13'

2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Explain why the two acute angles of a right triangle are complementary.
- Use $\sin \alpha = \cos(90^\circ - \alpha)$ and $\tan \alpha = \cot(90^\circ - \alpha)$.
- Convert a ratio of a large acute angle into one of a small angle.
- Simplify expressions that mix an angle with its complement.

TEXTBOOK

National	Tema 22, pp. 52–53
Cambridge	Extension beyond Stage 9
Workbook	—

01

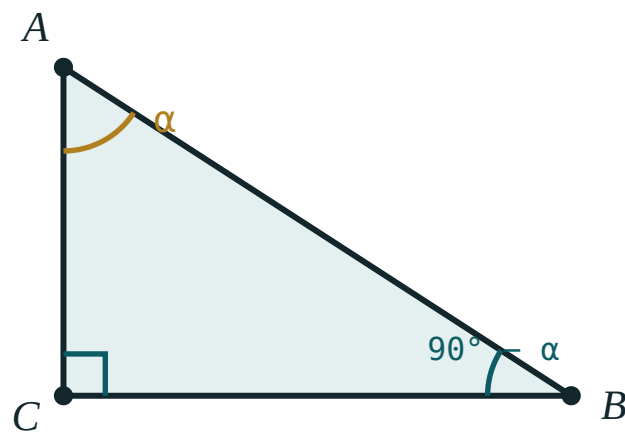
Explanation

Why the angles are complementary

In a right-angled triangle the three angles add to 180° and one of them is 90° . So the two acute angles add to 90° :

$$\alpha + \beta = 90^\circ, \text{ that is } \beta = 90^\circ - \alpha$$

Two angles adding to 90° are called **complementary**.



$$\begin{aligned}\sin \alpha &= \cos(90^\circ - \alpha) \\ \tan \alpha &= \cot(90^\circ - \alpha)\end{aligned}$$

The same two legs serve both angles — but their roles as “opposite” and “adjacent” are exchanged.

The exchange

Look at the figure. For the angle α , the leg a is opposite and b is adjacent. For the angle $\beta = 90^\circ - \alpha$, it is the other way round. So

$$\sin \alpha = \frac{a}{c} = \cos \beta = \cos(90^\circ - \alpha)$$

THE FOUR RELATIONS

$$\sin \alpha = \cos(90^\circ - \alpha) \quad \cdot \quad \cos \alpha = \sin(90^\circ - \alpha)$$

$$\tan \alpha = \cot(90^\circ - \alpha) \quad \cdot \quad \cot \alpha = \tan(90^\circ - \alpha)$$

Each function turns into its **co**-function.

That is where the names come from: “cosine” is short for *complementi sinus* — the sine of the complement.

What it is for

Old tables only listed angles up to 45° . Everything above that was read off the same page using these relations:

$$\sin 70^\circ = \cos 20^\circ \quad \cdot \quad \tan 80^\circ = \cot 10^\circ$$

Even with a calculator the relations are worth knowing, because they simplify expressions instantly:

$$\sin^2 40^\circ + \sin^2 50^\circ = \sin^2 40^\circ + \cos^2 40^\circ = 1$$

CHECK THE ANGLES REALLY ARE COMPLEMENTARY

$40^\circ + 50^\circ = 90^\circ$ ✓, so the swap is legal. For 40° and 60° it is not.

02 Worked examples

EXAMPLE 1 Write $\sin 63^\circ$ as a function of an angle below 45° .

- ① $90^\circ - 63^\circ = 27^\circ$
Find the complement.

- ② $\sin 63^\circ = \cos 27^\circ$
Sine becomes cosine.

ANSWER

$\cos 27^\circ$

EXAMPLE 2 Simplify $\sin^2 25^\circ + \sin^2 65^\circ$

① $25^\circ + 65^\circ = 90^\circ$
The angles are complementary.

② $\sin 65^\circ = \cos 25^\circ$

③ $= \sin^2 25^\circ + \cos^2 25^\circ$

④ $= 1$
The basic identity.

ANSWER**1****EXAMPLE 3** Simplify $\tan 35^\circ \cdot \tan 55^\circ$

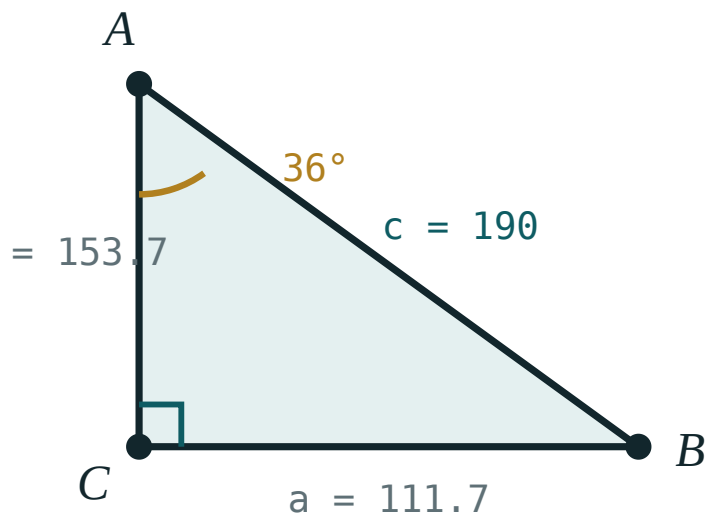
① $35^\circ + 55^\circ = 90^\circ$
Complementary.

② $\tan 55^\circ = \cot 35^\circ$

③ $\tan 35^\circ \cdot \cot 35^\circ = 1$
They are reciprocals.

ANSWER**1****03 Interactive model**

Set α to 30° and read both angles: the ratios at 30° and 60° are the same two numbers swapped.

α and $90^\circ - \alpha$ in one triangle α 36° β 54° a (opposite) **111.7**b (adjacent) **153.7**c (hyp) **190** $\sin \alpha$ **0.59** $\cos \alpha$ **0.81** $\tan \alpha$ **0.73** $a^2 + b^2$ **36100** c^2 **36100****04 Terminology**

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Complementary angles	To'ldiruvchi burchaklar	Дополнительные углы
Sum of angles	Burchaklar yig'indisi	Сумма углов
Co-function	Ko-funksiya	Кофункция
Convert	Aylantirish	Преобразовать
Acute angle	O'tkir burchak	Острый угол
Equal ratios	Teng nisbatlar	Равные отношения
Simplify	Soddalashtirish	Упростить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$\sin 70^\circ$ equals:

$\cos 70^\circ$

$\cos 20^\circ$

$\sin 20^\circ$

$\tan 70^\circ$

Two acute angles of a right triangle:

are equal

add to 90°

add to 180°

are unrelated

$\tan 35^\circ \cdot \tan 55^\circ$ equals:

0

1

2

$\tan 90^\circ$

$\sin^2 40^\circ + \sin^2 50^\circ$ equals:

1

2

0

$\sin^2 90^\circ$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Write $\sin 60^\circ$ as a cosine.

ANSWER $\cos 30^\circ$

2. Write $\cos 80^\circ$ as a sine.

ANSWER $\sin 10^\circ$

3. Write $\tan 70^\circ$ as a cotangent.

ANSWER $\cot 20^\circ$

4. Write $\cot 55^\circ$ as a tangent.

ANSWER $\tan 35^\circ$

5. Find the complement of 37° .

ANSWER 53°

6. $\alpha = 28^\circ$. Find β in a right triangle.

ANSWER 62°

7. $\sin 45^\circ$ and $\cos 45^\circ$ — are they equal?

ANSWER Yes, since 45° is its own complement.

Medium · the standard question

7 PROBLEMS

1. Write $\sin 63^\circ$ as a function of an angle below 45° .

ANSWER $\cos 27^\circ$

2. Write $\cos 18^\circ$ as a sine.

ANSWER $\sin 72^\circ$

3. Simplify $\sin^2 25^\circ + \sin^2 65^\circ$

ANSWER 1

4. Simplify $\tan 35^\circ \cdot \tan 55^\circ$

ANSWER 1

5. Simplify $\frac{\sin 40^\circ}{\cos 50^\circ}$

ANSWER 1

6. Simplify $\cos^2 15^\circ + \cos^2 75^\circ$

ANSWER 1

7. In a right triangle, $\sin \alpha = 0.6$. Find $\cos \beta$.

ANSWER 0.6 — since $\beta = 90^\circ - \alpha$

Hard · stretch and reason

7 PROBLEMS

1. Simplify $\sin^2 10^\circ + \sin^2 20^\circ + \sin^2 70^\circ + \sin^2 80^\circ$

ANSWER Pair 10° with 80° and 20° with 70° : $1 + 1 = 2$

2. Simplify $\tan 10^\circ \cdot \tan 20^\circ \cdot \tan 70^\circ \cdot \tan 80^\circ$

ANSWER Pair 10° with 80° and 20° with 70° : $1 \cdot 1 = 1$

3. Simplify $\frac{\sin 32^\circ}{\cos 58^\circ} + \frac{\cos 32^\circ}{\sin 58^\circ}$

ANSWER $1 + 1 = 2$

4. Prove that $\sin \alpha \cdot \cos(90^\circ - \alpha) + \cos \alpha \cdot \sin(90^\circ - \alpha) = 1$

ANSWER Both terms become $\sin^2 \alpha$ and $\cos^2 \alpha$.

5. If $\sin(2x) = \cos(3x)$ and both angles are acute, find x .

ANSWER $2x + 3x = 90^\circ$, so $x = 18^\circ$.

6. Arrange $\sin 20^\circ$, $\sin 50^\circ$, $\cos 20^\circ$, $\cos 50^\circ$ in increasing order.

ANSWER $\sin 20^\circ < \sin 50^\circ = \cos 40^\circ < \cos 20^\circ$; and $\cos 50^\circ = \sin 40^\circ$, so the order is $\sin 20^\circ < \cos 50^\circ < \sin 50^\circ < \cos 20^\circ$.

7. Simplify $(\sin 25^\circ + \cos 65^\circ)^2$

ANSWER $\cos 65^\circ = \sin 25^\circ$, so $(2 \sin 25^\circ)^2 = 4 \sin^2 25^\circ$

07 Homework

Homework — 5 problems

Geometry 8, Tema 22, pp. 52–53.

- Write $\sin 74^\circ$, $\cos 12^\circ$ and $\tan 66^\circ$ as functions of angles below 45° .
- Simplify $\sin^2 31^\circ + \sin^2 59^\circ$
- Simplify $\tan 22^\circ \cdot \tan 68^\circ$

4. Simplify $\frac{\cos 41^\circ}{\sin 49^\circ}$

5. If $\sin(3x) = \cos(x + 10^\circ)$ with both angles acute, find x .

The ratios for 30°, 45° and 60°

Three angles whose exact values come from half a square and half an equilateral triangle — and must be known by heart.

LESSON 26

1 × 40 MIN

GEOMETRY 8, TEMA 23

STAGE 9 · BEYOND

LESSON CLOCK

5'

12'

8'

13'

2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Derive the exact ratios for 45° from half a square.
- Derive the exact ratios for 30° and 60° from half an equilateral triangle.
- Recall the table of exact values.
- Use the exact values in calculations without a calculator.

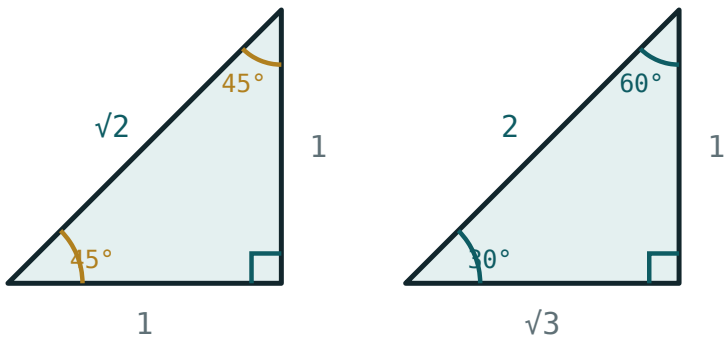
TEXTBOOK

National	Tema 23, pp. 54–55
Cambridge	Extension beyond Stage 9
Workbook	—

01

Explanation

The two triangles that give everything



Half a unit square gives 45°; half an equilateral triangle of side 2 gives 30° and 60°.

For 45°. Cut a square of side 1 along a diagonal. The two legs are 1 and 1, and the hypotenuse is $\sqrt{1+1} = \sqrt{2}$. So

$$\sin 45^\circ = \cos 45^\circ = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2} \quad \cdot \quad \tan 45^\circ = \cot 45^\circ = 1$$

For 30° and 60°. Cut an equilateral triangle of side 2 down its height. The base is halved to 1, the hypotenuse stays 2, and the height is $\sqrt{4-1} = \sqrt{3}$. The angles are 30° and 60°.

$$\sin 30^\circ = \frac{1}{2} \quad \cdot \quad \cos 30^\circ = \frac{\sqrt{3}}{2} \quad \cdot \quad \sin 60^\circ = \frac{\sqrt{3}}{2} \quad \cdot \quad \cos 60^\circ = \frac{1}{2}$$

The table

A	SIN A	COS A	TAN A	COT A
30°	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$	$\sqrt{3}$
45°	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1	1
60°	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$	$\frac{\sqrt{3}}{3}$

HOW TO REMEMBER IT

Write 1, 2, 3 under 30°, 45°, 60°, take the square root of each and divide by 2: $\frac{\sqrt{1}}{2} = \frac{1}{2}$, $\frac{\sqrt{2}}{2}$, $\frac{\sqrt{3}}{2}$.

That is the sine row. The cosine row is the same three numbers backwards.

Approximate values, for checking: $\sin 30^\circ = 0.5$, $\sin 45^\circ \approx 0.71$, $\sin 60^\circ \approx 0.87$.

Where they appear

These three angles run through the whole of school geometry: the height of an equilateral triangle ($h = a \cdot \sin 60^\circ = \frac{a\sqrt{3}}{2}$), the diagonal of a square ($d = a\sqrt{2}$), the

30° slope of a ramp, the shorter diagonal of a rhombus with a 60° angle.

02 Worked examples

EXAMPLE 1 Find $\sin 30^\circ + \cos 60^\circ$

$$\textcircled{1} \quad \sin 30^\circ = \frac{1}{2}$$

From the table.

$$\textcircled{2} \quad \cos 60^\circ = \frac{1}{2}$$

They are equal — 30° and 60° are complementary.

$$\textcircled{3} \quad \frac{1}{2} + \frac{1}{2} = 1$$

ANSWER

1

EXAMPLE 2 A right triangle has hypotenuse 10 and one angle 30° . Find both legs.

$$\textcircled{1} \quad a = c \cdot \sin 30^\circ = 10 \cdot \frac{1}{2} = 5$$

The side opposite the 30° .

$$\textcircled{2} \quad b = c \cdot \cos 30^\circ = 10 \cdot \frac{\sqrt{3}}{2} = 5\sqrt{3}$$

$$\textcircled{3} \quad 5\sqrt{3} \approx 8.66$$

Check: the shorter leg faces the smaller angle ✓

ANSWER

5 and $5\sqrt{3} \approx 8.66$

EXAMPLE 3 Simplify $2 \sin 30^\circ \cdot \cos 60^\circ + \tan^2 45^\circ$

$$\textcircled{1} \quad \sin 30^\circ = \cos 60^\circ = \frac{1}{2}$$

$$\textcircled{2} \quad 2 \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}$$

$$\textcircled{3} \quad \tan 45^\circ = 1, \text{ so } \tan^2 45^\circ = 1$$

$$\textcircled{4} \quad \frac{1}{2} + 1 = 1.5$$

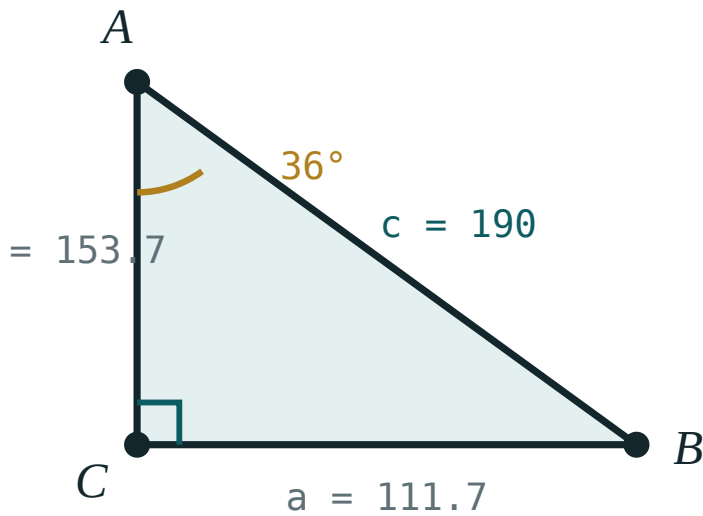
ANSWER

1.5

03 Interactive model

Set the angle to 30° , 45° and 60° in turn and compare the readouts with the table.

Check the table against the model



α 36°	β 54°	a (opposite) 111.7	b (adjacent) 153.7	c (hyp) 190	$\sin \alpha$ 0.59
$\cos \alpha$ 0.81	$\tan \alpha$ 0.73	$a^2 + b^2$ 36100	c^2 36100		

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Exact value	Aniq qiymat	Точное значение
Half of a square	Kvadratning yarmi	Половина квадрата
Equilateral triangle	Teng tomonli uchburchak	Равносторонний треугольник
Table of values	Qiyamatlar jadvali	Таблица значений
Special angle	Xos burchak	Специальный угол
Surd	Irratsional ildiz	Иррациональный корень
Derive	Keltirib chiqarish	Вывести
By heart	Yoddan	Наизусть

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$\sin 30^\circ$ equals:

$$\frac{1}{2}$$

$$\frac{\sqrt{3}}{2}$$

$$\frac{\sqrt{2}}{2}$$

$$1$$

$\cos 45^\circ$ equals:

$$\frac{1}{2}$$

$$\frac{\sqrt{2}}{2}$$

$$1$$

$$\sqrt{2}$$

$\tan 60^\circ$ equals:

$\sin 60^\circ$ equals:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\sin 30^\circ = ?$

ANSWER $\frac{1}{2}$

2. $\cos 60^\circ = ?$

ANSWER $\frac{1}{2}$

3. $\tan 45^\circ = ?$

ANSWER 1

4. $\sin 45^\circ = ?$

ANSWER $\frac{\sqrt{2}}{2}$

5. $\cos 30^\circ = ?$

ANSWER $\frac{\sqrt{3}}{2}$

6. $\tan 60^\circ = ?$

ANSWER $\sqrt{3}$

7. $\cot 45^\circ = ?$

ANSWER 1

Medium · the standard question

7 PROBLEMS

1. $\sin 30^\circ + \cos 60^\circ$

ANSWER 1

2. $\sin^2 45^\circ + \cos^2 45^\circ$

ANSWER 1

3. $\tan 30^\circ \cdot \tan 60^\circ$

ANSWER 1

4. A right triangle has hypotenuse 10 and an angle of 30° . Find both legs.

ANSWER 5 and $5\sqrt{3}$

5. A right triangle has hypotenuse 8 and an angle of 45° . Find both legs.

ANSWER $4\sqrt{2}$ each

6. $2 \sin 60^\circ = ?$

ANSWER $\sqrt{3}$

7. An equilateral triangle has side 6. Find its height.

ANSWER $3\sqrt{3}$

Hard · stretch and reason

7 PROBLEMS

1. Simplify $2 \sin 30^\circ \cdot \cos 60^\circ + \tan^2 45^\circ$

ANSWER 1.5

2. Simplify $\sin 60^\circ \cdot \cos 30^\circ + \sin 30^\circ \cdot \cos 60^\circ$

ANSWER $\frac{3}{4} + \frac{1}{4} = 1$

3. A ramp rises at 30° over a horizontal distance of 12 m. Find its height and length.

ANSWER $h = 12 \tan 30^\circ = 4\sqrt{3} \approx 6.93$, $\ell = 8\sqrt{3} \approx 13.86$

4. A rhombus has side 8 and an angle of 60° . Find its shorter diagonal.

ANSWER 8 — the triangle cut off is equilateral.

5. Find the exact area of an equilateral triangle of side a .

ANSWER $S = \frac{a^2\sqrt{3}}{4}$

6. Simplify $\frac{\sin 60^\circ}{\cos 60^\circ} - \tan 60^\circ$

ANSWER 0

7. A square has diagonal 10. Find its side exactly.

ANSWER $5\sqrt{2}$

07 Homework

Homework — 6 problems

Geometry 8, Tema 23, pp. 54–55. Learn the table before the next lesson.

- Write out the table of values for 30° , 45° and 60° from memory.
- Simplify $\sin 45^\circ \cdot \cos 45^\circ$
- A right triangle has hypotenuse 14 and an angle of 60° . Find both legs.

4. *An equilateral triangle has side 12. Find its height and area exactly.*
5. *Simplify $\tan 45^\circ + 2 \cos 60^\circ$*
6. *A square has side 7. Find its diagonal exactly and to 2 decimal places.*

The table of trigonometric values

Reading a ratio from an angle, and an angle from a ratio — on paper and on a calculator.

LESSON 27–28

2 × 40 MIN

GEOMETRY 8, TEMA 24

STAGE 9 · BEYOND

LESSON CLOCK

5'

12'

8'

13'

2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Read a value from a four-figure table or a calculator.
- Find the angle when the ratio is given (the inverse operation).
- Know how each ratio behaves as the angle grows from 0° to 90°.
- Estimate before calculating, and check the answer is sensible.

TEXTBOOK

National	Tema 24, pp. 56–57
Cambridge	Extension beyond Stage 9
Workbook	—

01

Explanation

Reading the table

A four-figure table gives $\sin$, $\cos$, $\tan$ and $\cot$ for every angle from 0° to 90° , usually in steps of $6'$ (six minutes of arc, one tenth of a degree). Find the row for the whole degrees and the column for the minutes.

On a calculator, make sure it is in **degree mode** (the display shows DEG). Then $\sin 37^\circ = 0.6018\dots$, $\cos 37^\circ = 0.7986\dots$, $\tan 37^\circ = 0.7536\dots$

CHECK THE MODE FIRST, EVERY TIME

In radian mode $\sin 37$ gives -0.6435 — a negative number, which is impossible for an acute angle. A negative sine is always a mode error.

The inverse direction

Given the ratio, find the angle. On the calculator this is the second function of the same key, written $\sin^{-1}$ or $\arcsin$:

$\sin \alpha = 0.5 \implies \alpha = 30^\circ \cdot \tan \alpha = 1 \implies \alpha = 45^\circ$

In a table you read backwards: find the value in the body and go out to the edge for the angle. If the exact value is not there, take the nearest — the answer is approximate anyway.

NOT A POWER

$\sin^{-1}x$ means “the angle whose sine is x ”. It is **not** $\frac{1}{\sin x}$.

How the ratios behave

A	0°	30°	45°	60°	90°	BEHAVIOUR
sin α	0	0.500	0.707	0.866	1	increases
cos α	1	0.866	0.707	0.500	0	decreases
tan α	0	0.577	1	1.732	—	increases without limit
cot α	—	1.732	1	0.577	0	decreases

Three facts to use as checks:

- $\sin$ and $\cos$ are always between 0 and 1;
- $\tan$ passes through 1 at 45° — below that it is less than 1, above it more;
- as α grows, $\sin$ and $\tan$ grow while $\cos$ and $\cot$ shrink.

02 Worked examples

EXAMPLE 1 Find $\sin 52^\circ$ and use it to find the leg opposite 52° in a triangle with hypotenuse 20.

① $\sin 52^\circ \approx 0.788$

From the table or calculator.

② $a = c \cdot \sin \alpha = 20 \cdot 0.788$

③ $a \approx 15.8$

Check: it is less than 20 ✓ and more than half of 20, as $52^\circ > 30^\circ$ ✓

ANSWER

≈ 15.8

EXAMPLE 2 Find the angle α if $\cos \alpha = 0.4226$.

① $\cos \alpha = 0.4226$

Given.

② Read the table backwards, or press $\cos^{-1}$.

③ $\alpha \approx 65^\circ$

Sensible: the cosine is under 0.5, so the angle is above 60° .

ANSWER

$\alpha \approx 65^\circ$

EXAMPLE 3 *Without a table, decide whether $\sin 40^\circ$ is bigger or smaller than $\cos 40^\circ$.*

① $40^\circ < 45^\circ$

② Below 45° the sine is the smaller of the two.
They are equal exactly at 45° .

③ $\sin 40^\circ < \cos 40^\circ$
Check: $0.643 < 0.766$ ✓

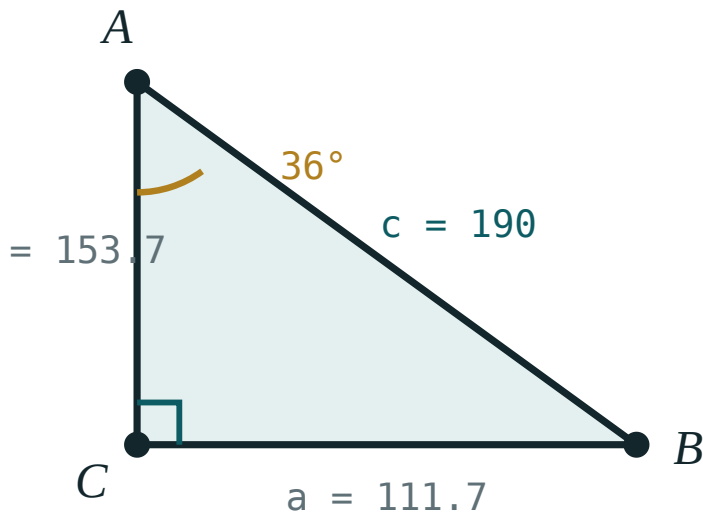
ANSWER

$$\sin 40^\circ < \cos 40^\circ$$

03 Interactive model

Sweep the angle from 10° to 80° and read the three ratios changing — sine up, cosine down.

How the ratios move with the angle



α 36°	β 54°	a (opposite) 111.7	b (adjacent) 153.7	c (hyp) 190	$\sin \alpha$ 0.59
$\cos \alpha$ 0.81	$\tan \alpha$ 0.73	$a^2 + b^2$ 36100	c^2 36100		

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Table of values	Qiymatlar jadvali	Таблица значений
Calculator	Kalkulyator	Калькулятор
Degree	Gradus	Градус
Minute (of arc)	Daqiqa	Минута
Inverse function	Teskari funksiya	Обратная функция
Increasing	O'suvchi	Возрастающая
Decreasing	Kamayuvchi	Убывающая
Round	Yaxlitlash	Округлить
Approximate value	Taqribiy qiymat	Приближённое значение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

As α grows from 0° to 90° , $\cos \alpha$:

increases

decreases

stays the same

increases then decreases

$\tan 50^\circ$ is:

less than 1

equal to 1

greater than 1

negative

$\sin^{-1}(0.5)$ equals:

2

30°

60°

0.5

A calculator gives $\sin 37 = -0.6435$. This means:

the angle is obtuse

the calculator is in radian mode

the sine can be negative

the answer is right

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Find $\sin 30^\circ$ from the table.

ANSWER 0.5

2. Find $\cos 60^\circ$.

ANSWER 0.5

3. Find $\tan 45^\circ$.

ANSWER 1

4. Find $\sin 50^\circ$ to 3 d.p.

ANSWER 0.766

5. Find $\cos 25^\circ$ to 3 d.p.

ANSWER 0.906

6. Find α if $\sin \alpha = 0.5$.

ANSWER 30°

7. Find α if $\tan \alpha = 1$.

ANSWER 45°

Medium · the standard question

7 PROBLEMS

1. Find $\sin 52^\circ$ to 3 d.p.

ANSWER 0.788

2. Find $\tan 68^\circ$ to 3 d.p.

ANSWER 2.475

3. Find α if $\cos \alpha = 0.4226$.

ANSWER 65°

4. Find α if $\tan \alpha = 0.5774$.

ANSWER 30°

5. A triangle has hypotenuse 20 and an angle of 52° . Find the opposite leg.

ANSWER ≈ 15.8

6. A triangle has hypotenuse 15 and an angle of 40° . Find the adjacent leg.

ANSWER ≈ 11.5

7. Which is bigger, $\sin 40^\circ$ or $\cos 40^\circ$?

ANSWER $\cos 40^\circ$

Hard · stretch and reason

7 PROBLEMS

1. Arrange in increasing order: $\sin 20^\circ$, $\sin 70^\circ$, $\cos 20^\circ$, $\cos 70^\circ$.

ANSWER $\cos 70^\circ = \sin 20^\circ < \sin 70^\circ = \cos 20^\circ$, so $\sin 20^\circ = \cos 70^\circ < \sin 70^\circ = \cos 20^\circ$

2. For which acute angles is $\tan \alpha > 1$?

ANSWER $45^\circ < \alpha < 90^\circ$

3. A calculator shows $\sin x = 1.2$. What has gone wrong?

ANSWER Impossible — the sine of an acute angle is under 1. A side has been divided the wrong way round.

4. A ramp has length 20 m and rises 7 m. Find its angle of slope.

ANSWER $\sin \alpha = 0.35$, so $\alpha \approx 20.5^\circ$

5. A tower is 45 m tall and its shadow is 30 m. Find the angle of elevation of the sun.

ANSWER $\tan \alpha = 1.5$, so $\alpha \approx 56.3^\circ$

6. Explain why $\tan \alpha$ has no value at 90° .

ANSWER The adjacent side would be 0, and division by 0 has no meaning.

7. Estimate $\sin 33^\circ$ without a table, using $\sin 30^\circ$ and $\sin 45^\circ$.

ANSWER Between 0.5 and 0.707, nearer the first — about 0.54. (True value 0.545.)

07 Homework

Homework — 5 problems

Geometry 8, Tema 24, pp. 56–57. Bring the table or a calculator in degree mode.

- Find $\sin 28^\circ$, $\cos 28^\circ$ and $\tan 28^\circ$ to 3 decimal places.
- Find α if $\sin \alpha = 0.8660$.
- Find α if $\cos \alpha = 0.7071$.
- A triangle has hypotenuse 25 and an angle of 36° . Find both legs.

5. Without a table, say which is bigger: $\tan 30^\circ$ or $\cot 30^\circ$. Explain.

Solving right-angled triangles

Given any two parts, find the other three — the skill the whole chapter has been building towards.

LESSON 29–30

2 × 40 MIN

GEOMETRY 8, ТЕМЫ 25–26

STAGE 9 · BEYOND

LESSON CLOCK

5'

12'

8'

13'

2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Solve a right triangle given two sides.
- Solve a right triangle given one side and one acute angle.
- Choose the ratio that uses the two parts you know.
- Solve practical problems of height, distance and slope.

TEXTBOOK

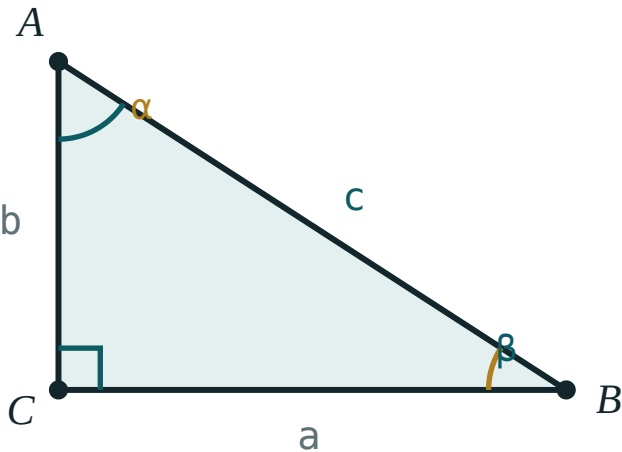
National	Темы 25–26, pp. 58–61
Cambridge	Extension beyond Stage 9
Workbook	—

01

Explanation

What “solving” means

A right triangle has six parts: three sides and three angles. One angle is already known (90°). Given any **two more** — as long as at least one is a side — the other three follow.



legs a, b · hypotenuse c

Six parts, one of them always 90° . Two more are enough to fix everything.

CHOOSING THE RATIO

Write down what you **know** and what you **want**, then pick the ratio containing exactly those two sides:

- opposite and hypotenuse $\rightarrow \sin$
- adjacent and hypotenuse $\rightarrow \cos$
- opposite and adjacent $\rightarrow \tan$ (or $\cot$)

Two sides and no angle? Use Pythagoras first, or a ratio to get the angle.

The two standard cases

Case 1 — two sides given. Find the third by Pythagoras, then an angle from a ratio, then the last angle by subtracting from 90° .

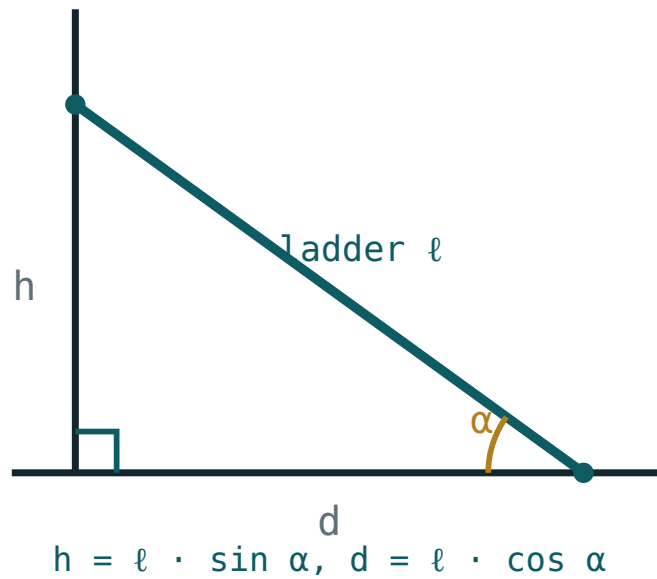
Case 2 — one side and one acute angle given. The other acute angle is $90^\circ - \alpha$ immediately. Then use the ratios to build the two missing sides:

$$a = c \cdot \sin \alpha \quad b = c \cdot \cos \alpha \quad a = b \cdot \tan \alpha$$

MULTIPLY OR DIVIDE?

Going from the hypotenuse to a leg you **multiply** by a ratio (the answer gets smaller). Going from a leg to the hypotenuse you **divide** (the answer gets bigger). If your hypotenuse comes out shorter than a leg, you divided the wrong way.

Practical problems



Every height-and-distance problem is this picture with different labels.

- **Angle of elevation** — measured *up* from the horizontal to the line of sight.
- **Angle of depression** — measured *down* from the horizontal.

They are alternate angles between two horizontals, so the angle of depression from the top of a tower equals the angle of elevation from the ground point. Draw the horizontal line at the observer's eye before anything else — most mistakes in these problems are mistakes in the diagram, not the trigonometry.

02 Worked examples**EXAMPLE 1** Solve the right triangle with $c = 20$ and $\alpha = 35^\circ$.

① $\beta = 90^\circ - 35^\circ = 55^\circ$

The other acute angle.

② $a = c \cdot \sin \alpha = 20 \cdot 0.5736 \approx 11.5$

Opposite and hypotenuse $\rightarrow$ sine.

③ $b = c \cdot \cos \alpha = 20 \cdot 0.8192 \approx 16.4$

Adjacent and hypotenuse $\rightarrow$ cosine.

④ check: $11.5^2 + 16.4^2 \approx 400 \checkmark$

Pythagoras confirms it.

ANSWER

$$\beta = 55^\circ, a \approx 11.5, b \approx 16.4$$

EXAMPLE 2 *Solve the right triangle with legs $a = 5$ and $b = 12$.*

$$\textcircled{1} \quad c = \sqrt{25 + 144} = 13$$

Pythagoras.

$$\textcircled{2} \quad \tan \alpha = \frac{5}{12} \approx 0.4167$$

Two legs $\rightarrow$ tangent.

$$\textcircled{3} \quad \alpha \approx 22.6^\circ$$

Inverse tangent.

$$\textcircled{4} \quad \beta \approx 67.4^\circ$$

$$90^\circ - 22.6^\circ$$

ANSWER

$$c = 13, \alpha \approx 22.6^\circ, \beta \approx 67.4^\circ$$

EXAMPLE
3**From a point 40 m from the foot of a tower the angle of elevation of its top is 38° . Find the height.**

- ① Draw the tower, the ground and the line of sight.

The tower is opposite the 38° angle.

② $\tan 38^\circ = \frac{h}{40}$

Opposite and adjacent $\rightarrow$ tangent.

③ $h = 40 \cdot \tan 38^\circ = 40 \cdot 0.7813$

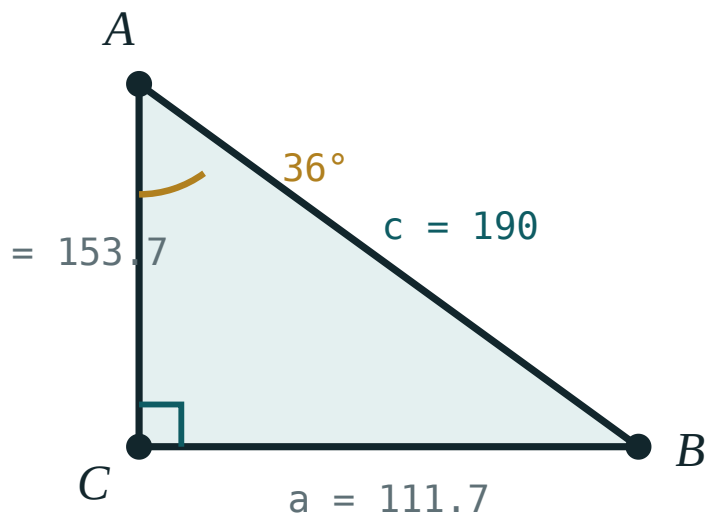
④ $h \approx 31.3 \text{ m}$

Sensible: less than the 40 m distance, since $38^\circ < 45^\circ$ ✓

ANSWER $\approx 31.3 \text{ m}$ **03 Interactive model**

Give the class two parts, let them predict the other three, then check with the model.

Solve the right triangle

 α 36° β 54° a (opposite) **111.7**b (adjacent) **153.7**c (hyp) **190** $\sin \alpha$ **0.59** $\cos \alpha$ **0.81** $\tan \alpha$ **0.73** $a^2 + b^2$ **36100** c^2 **36100****04 Terminology**

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Solve a triangle	Uchburchakni yechish	Решить треугольник
Given	Berilgan	Дано
Find	Topish kerak	Найти
Angle of elevation	Ko'tarilish burchagi	Угол возвышения
Angle of depression	Pasayish burchagi	Угол понижения
Slope	Qiyalik	Уклон
Height	Balandlik	Высота
Horizontal distance	Gorizontal masofa	Горизонтальное расстояние
Accuracy	Aniqlik	Точность

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Given the hypotenuse and the angle, the opposite leg is:

$$c \cdot \cos \alpha$$

$$c \cdot \sin \alpha$$

$$\frac{c}{\sin \alpha}$$

$$c \cdot \tan \alpha$$

Given both legs, to find an angle use:

$$\sin$$

$$\cos$$

$$\tan$$

$$\text{Pythagoras}$$

A leg is 6 and the angle opposite it is 30° . The hypotenuse is:

$$3$$

$$12$$

$$6\sqrt{3}$$

$$2$$

The angle of depression from a tower top equals:

the angle of elevation from the ground point

half of it

its complement

nothing in particular

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $c = 10, \alpha = 30^\circ$. Find a .

ANSWER 5

2. $c = 10, \alpha = 30^\circ$. Find β .

ANSWER 60°

3. $c = 8, \alpha = 45^\circ$. Find a .

ANSWER $4\sqrt{2} \approx 5.66$

4. $a = 3, b = 4$. Find c .

ANSWER 5

5. $a = 6, b = 6$. Find α .

ANSWER 45°

6. $c = 20, \alpha = 60^\circ$. Find b .

ANSWER 10

7. $\alpha = 25^\circ$. Find β .

ANSWER 65°

Medium · the standard question

7 PROBLEMS

1. $c = 20, \alpha = 35^\circ$. Solve the triangle.

ANSWER $\beta = 55^\circ, a \approx 11.5, b \approx 16.4$

2. $a = 5, b = 12$. Solve the triangle.

ANSWER $c = 13, \alpha \approx 22.6^\circ, \beta \approx 67.4^\circ$

3. $a = 7, c = 25$. Find b and α .

ANSWER $b = 24, \alpha \approx 16.3^\circ$

4. $b = 15, \alpha = 40^\circ$. Find a .

ANSWER $a = 15 \tan 40^\circ \approx 12.6$

5. $a = 9, \alpha = 30^\circ$. Find c .

ANSWER 18

6. From 40 m away the angle of elevation of a tower top is 38° . Find its height.

ANSWER ≈ 31.3 m

7. A ladder 6 m long makes 65° with the ground. How high does it reach?

ANSWER ≈ 5.4 m

Hard · stretch and reason

7 PROBLEMS

1. A ramp 15 m long rises 4 m. Find its angle of slope and its horizontal run.

ANSWER $\sin \alpha = 0.267$, $\alpha \approx 15.5^\circ$, run ≈ 14.5 m

2. From the top of a 50 m cliff the angle of depression of a boat is 22° . How far is the boat from the foot?

ANSWER $\frac{50}{\tan 22^\circ} \approx 123.8$ m

3. An isosceles triangle has equal sides 10 and apex angle 40° . Find its base.

ANSWER Half-base $10 \sin 20^\circ \approx 3.42$, so the base is ≈ 6.84 .

4. A rhombus has side 12 and an angle of 70° . Find both diagonals.

ANSWER $d_1 = 24 \sin 35^\circ \approx 13.8$, $d_2 = 24 \cos 35^\circ \approx 19.7$

5. Two people stand 100 m apart on opposite sides of a tower and see its top at 30° and 45° . Find the height.

ANSWER $h \cot 30^\circ + h \cot 45^\circ = 100$, so $h \approx 36.6$ m

6. A right triangle has area 24 and one leg 6. Find its angles.

ANSWER other leg 8, $\tan \alpha = \frac{8}{6}$, $\alpha \approx 53.1^\circ$, $\beta \approx 36.9^\circ$

7. A regular hexagon has side 6. Find the distance between two opposite sides.

ANSWER $2 \cdot 6 \sin 60^\circ = 6\sqrt{3} \approx 10.39$

67 Homework

Homework — 6 problems

Geometry 8, Темы 25–26, pp. 58–61. Draw and label a figure for every question.

- $c = 30$, $\alpha = 25^\circ$. Solve the triangle.
- $a = 8$, $b = 15$. Solve the triangle.

3. $a = 12$, $\alpha = 60^\circ$. Find c and b .
4. A ladder 8 m long makes 70° with the ground. How high does it reach?
5. From 60 m away the angle of elevation of a mast top is 42° . Find its height.
6. A ramp 20 m long rises 5 m. Find its angle of slope.

Practical tasks

— bearings and scale drawings

The national “practical tasks” lesson, carrying the Cambridge Stage 9 topic of bearings.

LESSON 31

1 × 40 MIN

GEOMETRY 8, TEMA 28

STAGE 9 · 13.1 [CAMBRIDGE INSERT]

LESSON CLOCK

5'

12'

8'

13'

2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Measure and write a bearing as three figures from north, clockwise.
- Find a back bearing.
- Make and read a scale drawing.
- Combine bearings with right-angle trigonometry to find a distance.

TEXTBOOK

National	Tema 28, pp. 64–66
Cambridge	Learner’s Book pp. 271–277
Workbook	Workbook 13.1

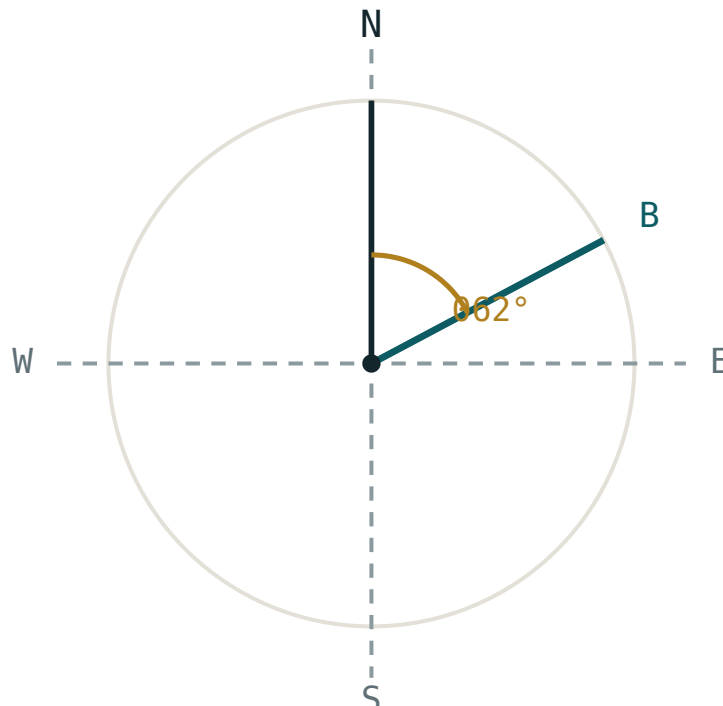
01

Explanation

What a bearing is

DEFINITION

A **bearing** is an angle measured **from north, clockwise**, and written with **three figures**.



The bearing of B from A is 062° — measured from north, turning clockwise.

All three parts matter. 62° must be written 062° ; a bearing measured anticlockwise or from another direction is simply wrong. The four cardinal directions are 000° (N), 090° (E), 180° (S) and 270° (W).

“THE BEARING OF B FROM A”

Stand at A, face north, and turn clockwise until you are looking at B. The point you stand at is the one after the word “from”.

Back bearings

The bearing of A from B is the **back bearing**. Because the two north lines are parallel, the two bearings differ by exactly 180° :

$$\text{back bearing} = \text{bearing} + 180^\circ \quad (\text{or } - 180^\circ \text{ if that exceeds } 360^\circ)$$

Bearing $062^\circ \rightarrow$ back bearing 242° . Bearing $310^\circ \rightarrow$ back bearing 130° .

Scale drawings

A scale of $1 : 50\,000$ means 1 cm on the paper is 50 000 cm — that is 500 m — on the ground. To make a scale drawing:

1. Draw a north line at the starting point.
2. Measure the bearing with a protractor, clockwise from north.

3. Convert the real distance to a paper distance and mark it off.

4. Repeat from the new point, with a fresh north line.

Then measure the answer off the drawing and convert back. A scale drawing gives a good enough answer for most practical work — and it is a check on the trigonometry.

Bearings with trigonometry

When the bearings are 090° apart, the path makes a right angle and the whole of this chapter applies.

Example. A ship sails 8 km on a bearing of 000° then 6 km on 090° . The two legs are perpendicular, so the distance from the start is $\sqrt{64 + 36} = 10$ km, and its bearing

from the start is $\tan^{-1} \frac{6}{8} \approx 037^\circ$.

02 Worked examples

EXAMPLE 1 The bearing of B from A is 145° . Find the bearing of A from B.

① $145^\circ + 180^\circ = 325^\circ$
Add 180° for the back bearing.

② $325^\circ < 360^\circ$
so no adjustment is needed.

③ 325°

ANSWER

325°

EXAMPLE 2 A ship sails 12 km due north, then 5 km due east. Find its distance and bearing from the start.

① North then east — the two legs are perpendicular.

② $d = \sqrt{144 + 25} = 13 \text{ km}$

Pythagoras.

③ $\tan \alpha = \frac{5}{12} \approx 0.4167$

East over north.

④ $\alpha \approx 22.6^\circ$, so the bearing is 023°

Measured clockwise from north.

ANSWER

13 km on a bearing of 023°

EXAMPLE 3 On a map of scale 1 : 25 000 two towns are 7.2 cm apart. Find the real distance.

① $1 \text{ cm} \rightarrow 25\,000 \text{ cm}$

The scale.

② $25\,000 \text{ cm} = 250 \text{ m}$

Convert to metres.

③ $7.2 \cdot 250 = 1800 \text{ m}$

④ $= 1.8 \text{ km}$

ANSWER

1.8 km

03 Interactive model

Have a learner walk a bearing across the classroom while the class calls out the angle.

Bearings, step by step

Bearing 062° .

1 The two north lines are parallel, so the bearings differ by 180° .

2 $062^\circ + 180^\circ = 242^\circ$

3 Under 360° , so no adjustment.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Bearing	Azimut	Азимут
North	Shimol	Север
Clockwise	Soat yo'nalishi bo'yicha	По часовой стрелке
Back bearing	Teskari azimut	Обратный азимут
Scale	Masshtab	Масштаб
Scale drawing	Masshtabli chizma	Чертёж в масштабе
Compass	Kompas	Компас
Due east	Aniq sharq	Точно на восток
Protractor	Transportir	Транспортир

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A bearing is measured:

from east, anticlockwise

from north, clockwise

from north, anticlockwise

from the horizontal

Due west as a bearing is:

090°

180°

270°

360°

The bearing of B from A is 100°. The bearing of A from B is:

080°

200°

260°

280°

On a 1 : 50 000 map, 1 cm represents:

50 m

500 m

5 km

50 km

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Write due north as a bearing.

ANSWER 000°

2. Write due east as a bearing.

ANSWER 090°

3. Write due south as a bearing.

ANSWER 180°

4. Write due west as a bearing.

ANSWER 270°

5. Write 45° north-east as a three-figure bearing.

ANSWER 045°

6. The bearing of B from A is 070° . Find the back bearing.

ANSWER 250°

7. On a $1 : 100\,000$ map, 1 cm represents how far?

ANSWER 1 km

Medium · the standard question

7 PROBLEMS

1. *The bearing of B from A is 145° . Find the bearing of A from B.*

ANSWER 325°

2. *The bearing of B from A is 310° . Find the back bearing.*

ANSWER 130°

3. *On a $1 : 25\,000$ map two towns are 7.2 cm apart. Find the real distance.*

ANSWER 1.8 km

4. *A ship sails 12 km north then 5 km east. Find its distance from the start.*

ANSWER 13 km

5. *The same ship: find its bearing from the start.*

ANSWER 023°

6. *A real distance of 4 km on a $1 : 50\,000$ map is how many cm ?*

ANSWER 8 cm

7. *The bearing of B from A is 205° . Find the back bearing.*

ANSWER 025°

Hard · stretch and reason

7 PROBLEMS

1. A ship sails 9 km on 000° , then 12 km on 090° . Find its distance and bearing from the start.

ANSWER 15 km on 053°

2. A walker goes 5 km on 090° then 5 km on 180° . Find the distance and bearing from the start.

ANSWER $5\sqrt{2} \approx 7.07$ km on 135°

3. The bearing of B from A is 062° and $AB = 20$ km. How far north and how far east is B?

ANSWER $20 \cos 62^\circ \approx 9.4$ km north, $20 \sin 62^\circ \approx 17.7$ km east

4. A plane flies 100 km on a bearing of 030° . Find its change in latitude (north) and longitude (east) in km.

ANSWER 86.6 km north, 50 km east

5. Two ports are 40 km apart, one due east of the other. A ship is 30 km north of the western port. Find its bearing from the eastern port.

ANSWER $\tan \alpha = \frac{40}{30}$, so the bearing is $360^\circ - 53.1^\circ = 307^\circ$.

6. Explain why a back bearing is exactly 180° from the original.

ANSWER The two north lines are parallel, so the bearings are co-interior angles adding to a straight-line turn.

7. A map has scale 1 : 20 000. A lake measures 3 cm by 4 cm on it. Find its real area in km^2 .

ANSWER $600 \text{ m} \times 800 \text{ m} = 0.48 \text{ km}^2$

07 Homework

Homework — 5 problems

Geometry 8, Tema 28, pp. 64–66, and Cambridge Learner's Book 13.1. Bring a protractor.

- Write these as three-figure bearings: north-east, south-west, due south.
- The bearing of B from A is 118° . Find the bearing of A from B.

3. *On a 1 : 50 000 map two villages are 5.6 cm apart. Find the real distance.*
4. *A ship sails 8 km north then 15 km east. Find its distance and bearing from the start.*
5. *Make a scale drawing (1 cm : 1 km) of a journey 6 km on 070° followed by 4 km on 160°, and measure the direct distance home.*

Control work

2 · Pythagoras and trigonometry

The Chapter II assessment, and the four errors it produces every year.

LESSON 32

1 × 40 MIN

GEOMETRY 8, ТЕМЫ 29–30

STAGE 9 · UNIT 5 CHECK

LESSON CLOCK

2'

36'

2'

Setting up

2 min

The paper

36 min

Collect in

2 min

LEARNING OBJECTIVES

- Assess Pythagoras, the four ratios, the identity, special angles and solving triangles.
- Produce a labelled figure and a stated reason for every answer.
- Diagnose each lost mark by error type.

TEXTBOOK

National

pp. 67–68

Cambridge

Learner’s Book
pp. 124–126

Workbook

Workbook 5.5

01

Explanation

The paper — 40 minutes, 7 tasks, 14 marks

TASK	TOPIC	SOURCE
1	Find a side by Pythagoras	Тема 17
2	Test a triangle for a right angle	Тема 18
3	Find the four ratios from the sides	Тема 15
4	Use the basic identity	Темы 20–21
5	Exact value for 30°, 45° or 60°	Тема 23
6	Solve a right triangle	Темы 25–26
7	A practical height or distance problem	Тема 28

Say before the paper starts: a numerical answer with no figure and no reason earns one mark out of two. Round to one decimal place unless an exact value is asked for.

Work on mistakes — the four errors

WHAT GOES WRONG

1. **Added where a subtraction was needed** — using $c^2 = a^2 + b^2$ when looking for a leg.
2. **Opposite and adjacent swapped** — sine used where cosine was needed.
3. **The calculator in radian mode**, giving a negative or absurd ratio.
4. **Multiplied instead of divided** when going from a leg to the hypotenuse.

Each has a one-second check: the hypotenuse must be the longest side; a sine must be under 1; the answer must be sensible on the figure. The Hard set below is built from these four.

02 Worked examples

EXAMPLE 1 Find the error: hypotenuse 13, one leg 5, so the other leg is $\sqrt{169 + 25} \approx 13.9$

- 1 The answer is longer than the hypotenuse — impossible.
That check alone catches it.

- 2 Looking for a **leg**, so subtract.

- 3 $b = \sqrt{169 - 25} = \sqrt{144} = 12$

ANSWER

12

EXAMPLE 2 Find the error: a leg is 6 and the angle opposite it is 30° , so the hypotenuse is $6 \cdot \sin 30^\circ = 3$

① The hypotenuse came out shorter than the leg — impossible.

② $\sin 30^\circ = \frac{a}{c}$, so $c = \frac{a}{\sin 30^\circ}$

Going leg $\rightarrow$ hypotenuse means dividing.

③ $c = \frac{6}{0.5} = 12$

ANSWER

12

03 Interactive model

In the mistakes lesson: show the wrong working, take a vote on the error type, then reveal.

Diagnose the error

Claimed: hypotenuse 13, leg 5, other leg $\sqrt{169 + 25} \approx 13.9$

- ① A leg cannot be longer than the hypotenuse.

The
instant
check.

- ② To find a leg, subtract:

$$\begin{array}{r} b^2 \\ = c^2 \\ - \\ a^2 \end{array} .$$

- ③ $b = \sqrt{144} = 12$

And
5,
12,
13
is a
familiar
triple.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Solve a triangle	Uchburchakni yechish	Решить треугольник
Figure	Chizma	Чертёж
Reason	Asos	Обоснование
Round to	Yaxlitlash	Округлить до
Error	Xato	Ошибка
Exact value	Aniq qiymat	Точное значение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Hypotenuse 25, one leg 7. The other leg is:

Given the hypotenuse and the angle, the adjacent leg is:

A calculator returns $\sin 40 = 0.745$. This is:

A leg is 9 and the angle opposite is 30° . The hypotenuse is:

4.5

18

$9\sqrt{3}$

15

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. **Variant 1, task 1.** Legs 12 and 16. Find the hypotenuse.

ANSWER 20

2. **Task 2.** Is 7, 24, 25 right-angled?

ANSWER Yes — $49 + 576 = 625$.

3. **Task 3.** $a = 6$, $b = 8$, $c = 10$. Find $\sin \alpha$ and $\cos \alpha$.

ANSWER 0.6 and 0.8

4. **Task 4.** $\sin \alpha = 0.6$. Find $\cos \alpha$.

ANSWER 0.8

5. **Task 5.** Find $\sin 60^\circ$ exactly.

ANSWER $\frac{\sqrt{3}}{2}$

6. **Task 6.** $c = 10$, $\alpha = 30^\circ$. Find both legs.

ANSWER 5 and $5\sqrt{3}$

7. **Task 7.** A ladder 10 m long has its foot 6 m from a wall. How high does it reach?

ANSWER 8 m

Medium · the standard question

7 PROBLEMS

1. **Variant 2, task 1.** Hypotenuse 26, one leg 24. Find the other leg.

ANSWER 10

2. **Task 2.** Classify the triangle with sides 6, 7, 10.

ANSWER $36 + 49 = 85 < 100$ — obtuse.

3. **Task 3.** $a = 9$, $b = 12$, $c = 15$. Find $\tan \alpha$ and $\cot \alpha$.

ANSWER 0.75 and $\frac{4}{3}$

4. **Task 4.** $\cos \alpha = \frac{7}{25}$. Find $\sin \alpha$.

ANSWER $\frac{24}{25}$

5. **Task 5.** Find $\tan 30^\circ$ exactly.

ANSWER $\frac{\sqrt{3}}{3}$

6. **Task 6.** $a = 5$, $b = 12$. Solve the triangle.

ANSWER $c = 13$, $\alpha \approx 22.6^\circ$, $\beta \approx 67.4^\circ$

7. **Task 7.** From 50 m away the angle of elevation of a tower top is 40° . Find its height.

ANSWER ≈ 42.0 m

Hard · stretch and reason

7 PROBLEMS

1. Find the error: hypotenuse 13, leg 5, other leg $\sqrt{169 + 25}$

ANSWER Added instead of subtracted. Correct: 12.

2. Find the error: $c = 20$, $\alpha = 35^\circ$, adjacent leg $= 20 \sin 35^\circ$

ANSWER Sine uses the opposite leg. Correct: $20 \cos 35^\circ \approx 16.4$.

3. Find the error: $\sin 37^\circ = -0.6435$

ANSWER The calculator is in radian mode. Correct: ≈ 0.602 .

4. Find the error: leg 6 opposite 30° , so $c = 6 \sin 30^\circ = 3$

ANSWER Leg $\rightarrow$ hypotenuse means dividing. Correct: 12.

5. Find the error: $\sin \alpha = 1.25$ for an acute angle

ANSWER Impossible — a sine is under 1. A side has been divided the wrong way round.

6. Find the error: “sides 5, 12, 13 give $\sin \alpha = \frac{13}{5}$ ”

ANSWER The hypotenuse belongs below. Correct: $\sin \alpha = \frac{5}{13}$.

7. Find the error: “ $\tan 45^\circ = 0$, since the legs are equal”

ANSWER Equal legs give $\frac{a}{a} = 1$, not 0.

07 Homework

After the control work

Re-solve every task you lost marks on. Quarter III opens with the coordinate method.

1. Re-solve every task you lost marks on, with a labelled figure and a stated reason.
2. Write out the four errors from the Hard set in your own words.

3. *Learn the table of exact values for 30° , 45° and 60° — Quarter III assumes it.*